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ABSTRACT

Numerical Methods for Phase-field Model and its Applications to Multi-phase Flow

Numerical methods for solving partial differential equations (PDEs) were used first by John von Neumann in the mid of 1940s as an effective tool. Since then, numerical computing has become the most versatile tool to experiments and complement theory. In principle, all mathematical equations can be easily solved numerically in comparison to the theoretical approach. They are low cost, high efficiency and no danger when compared with experimental approach. The driving force of numerical methods comes from practical application in all branches of science, engineering and other disciplines. The starting point of computational methods is a mathematical model, the form and origin of which depends on the particular field of study.

There exist many important physical and biological processes in nature that can be represented by mathematical models. However, a physical and mathematical interpretation of the models and their numerical solutions is always a vital part of the computational science. Although, analytical solutions are difficult to obtain for many complex phenomena governed by nonlinear PDEs. However, with the rapid development in computational techniques and availability of high-speed computers, there is a continuously widening scope of nonlinear problems that can be solved numerically. Appropriate numerical algorithms, in particular those for solving time-dependent nonlinear PDEs are in heart of many of advanced scientific computations and software development.

Moving free boundary problems are present in nature and many areas of physical and biological sciences. Examples include impact of a droplet on a solid surface, image segmentation, surface waves, jet breakup, realistic interfaces in animation movies, crystal and tumor growth and many others where the simulation of moving interfaces plays a key role in the problem to be solved. In dealing with moving boundary problems, an important consideration is how to model the moving boundary or interfacial surface on which the boundary conditions are imposed. Mathematical models adopted both in analytical and numerical studies for variety of free boundary problems are classified into two types, sharp interface and diffuse interface models. Sharp interface models like (level set method) assumes that the interface has zero thickness. However, in

phase transition, the existence of transition zone introduced an idea of diffuse interface by Gibbs, which allow the interface to have finite thickness. A type of diffuse interface model with particular interest is phase-field based model by the introduction of an order-variable representing the interface. In such approach the phase-field variable is continuous as a function of space and time.

Nevertheless, the partial differential equations describing the two-phase flow are highly nonlinear and numerical simulation is often necessary to solve them. This is why phase-field methods are numerically attractive with no tracking of interface explicitly but can be obtained as a part of the solution processes.

This work deals with the development of numerical methods for solving phase-field models with some real world applications. There are several discretization method like finite difference method, finite volume method and finite element method etc. We have adapted the conforming finite element method for spatial discretization and have used different diagonally implicit schemes for time discretization. The performance of the proposed numerical algorithms in term of their accuracy and CPU time are demonstrated. The comparisons with analytical, experimental and numerical results are also provided for validation and verification of the computed results. The numerical simulations were carried out using DUNE-PDELab, which is a software tool for solving partial differential equations.

This thesis is organized as follows: In chapter 1, an introduction of the phase-field models with some fundamental aspects and applications are given. Chapter 2 provides numerical methods with some discretization techniques and their modifications. Chapter 3 is based on the development of an efficient time stepping scheme for solving of 2nd-order nonlinear Allen-Cahn equation. Error estimates at different degrees of freedom are also provided using available exact solution. In chapter 4, the method developed in chapter 2 is further extended to solve 4th-order Cahn-Hillard equation with variable mobility. In chapter 5, a system of Allen-Cahn equation coupled with heat equation is solved with its application to dendritic crystal growth phenomena. Chapter 6 is based on the mathematical modeling and its numerical simulation of tumor hypoxia targeting in cancer treatment. Chapter 7 concludes this work with some future research directions.

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LIST OF ABBREVIATIONS

FDM	Finite difference method
FEM	Finite element method
PDEs	Partial differential equations
SCN	Succinonitrile
DNA	Deoxyribonucleic acid
TOL	Tolerance
ODEs	Ordinary differential equations
CPU	Central processing unit
DIRK	Diagonally implicit Runge-Kutta
DIFST	Diagonally implicit fractional step θ
GHz	Gigahertz
RAM	Random access memory
DOF	Degrees of freedom
DUNE	Distributed and unified numerics environment
MAX L-IT	Maximum linear iterations
MAX NL-IT	Maximum nonlinear iterations
LS TIME	Linear solver time
BiCG	Bi-Conjugate gradient

Chapter 1

Introduction

This chapter deals with an introduction of phase-field models and some of its applications. We begin with physical and mathematical background and discuss the gradient flows represented by the convection diffusion equations. Challenges in modeling and numerical simulation are discussed. An overview of some scientific problems that can be solved using phase-field models are also given.

1.1 Phase-field Modeling

Several physical processes in nature are governed by ordinary/partial differential equations (ODEs/PDEs). So, the knowledge of mathematical properties and solutions to these models are also required. Sophisticated numerical methods, especially for the solution of nonlinear PDEs, are at the heart of many of these advanced scientific computing. In fact, numerical computations is not only a research tool, but it changes the nature of experiments and scope of theory. For example; simulations of nuclear explosions replace soil investigations and understanding of the chaotic dynamics and fractal behavior can be gained only by the repetition of numerical experiments [1].

Free boundary problems are present in nature and in many areas of physical and biological sciences. In dealing with moving boundary problems, an important consideration is how to model the moving boundary or interfacial surface on which the boundary conditions are imposed [2, 3]. The partial differential equations describing the interfacial phenomena are nonlinear and stiff. Also, the boundary conditions at the moving interfaces require a particular treatment which is difficult and often tedious. So it is difficult to find an exact solution of such problem and numerical simulation is an alternative. Mathematical models used for free boundary phenomena are classified into two types, sharp and diffuse interface models. Sharp interface models assumes that the interface has zero thickness. However, in phase transition, the existence of transition allows the interface to have finite thickness [4]. A type of diffuse interface model with particular interest is phase-field based model by an introduction of a phase-field function. The phase-field function is assumed to take predetermined values in each phase and continuously varying values at the transition region (diffuse interface) as shown in figure 1.1(a).

In recent years, phase-field models are gaining popularity in scientific computing community due to their wide range of applications in science and engineering. Problems

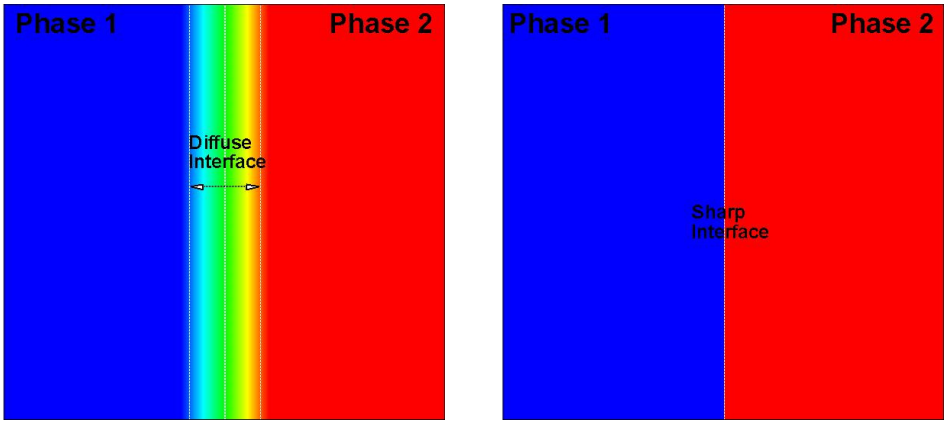


Figure 1.1: (a) Schematic diagram of (a) diffuse interface (b) sharp interface

include microstructure formation in solidification process [5], on solid state transformation [6], multi-phase flow [7], Hele-Shaw instabilities, Marangoni convection, droplet and vesicle dynamics, image in-painting [8] and polymer blend [9].

All these problems include the movement of interfaces or boundaries in response to a coupling from the boundary with one or more transportation procedure (e.g stress, temperature, shear flows, diffusion etc). This interaction produces morphological instability and discontinuity. Graphically, this discontinuity having sharp interface between two phases is illustrated in figure 1.1(b). For example, by taking vapor-liquid interface, the quantity which distinguishes the two states (i.e. vapor and liquid state) is the deference of molecules density, which is comparatively low in the vapor phase than the liquid phase. The van der Waals theory was the first example of the liquid-vapor interface [10]. All such type of models having diffuse interfaces with the true intrinsic thickness of the physical interface are used in phase separation phenomena. For two-phase flows, by considering a homogeneous assimilation of two immiscible liquids within any size of the domain of two liquids under normal conditions, the interface have usually very small size. Conversely, in case of polymer blends, the width of the diffuse interface between the two liquids is of the order of magnitude of size of molecules. Therefore, the interfacial thickness is very small as compared to domain size [11].

In sharp interface formulation, the Dirac distribution which describe the capillary forces due to the surface tension is located near the interface causes discontinuities/singularities. This type of formulation for sharp interface can be (and has been) utilized accurately to perform numerical simulations of two-phase/multi-phase flows [12]. However, the phase-field modeling, an obvious idea is to make it more suitable to overcome the sin-

gularities, i.e., to replace the step and Dirac delta function by smooth profile. These singularities/thickness are well-defined mathematically and lead to non-zero length scale (say ε). In recent years, sharp-interface problems were simulated by choosing the lager value of interfacial thickness ε by Karma and Rappel [13] and Echebarria et al. [14]. So the maximum value of thickness ε may be chosen but must be smaller than the length scale of two-phase flow pattern to define geometry for smoothness of the problem. In addition, thermodynamics directly relates with equations of motion and mathematical modeling in such a way are rarely being used for practical applications due to the separation of length scales. If physically, the diffuse interface has same order of interfacial thickness, needed by equations need to discretize on microscopic level for resolving the interfacial profile smoothly. So the time and length scales restricts severely in the numerical simulations of phase-field models and these restrictions (very thicker interfaces than the physical ones) can be handled by choosing larger system and small time step size. In the phase-field model, smaller interficial thickness can be replaced by finite width for which numerical simulations are reliable.

The phase-field models are mathematical representation of problems having interfacial dynamics. There are variety of industrial applications of these models like viscous fingering [15], fracture dynamics [16], vesicle dynamics [17], tumor growth [18], impact of a droplet on a solid surface [19], drop coalescence [20], realistic interfaces in computer graphics and animation movies [21], and understanding the material properties at micro-structural scales [22] etc. In the phase-field models, boundary conditions are substituted at the interfaces by a partial differential equation (PDE) to evaluate order parameter of the auxiliary field (phase-field). The phase variable $u(\mathbf{x},t)$ varies smoothly from one fixed value (for example $u = 1$) in one phase to another fixed value ($u = -1$) in other phase with a diffusive region ε near the interface as shown in figure 1.2. Also, the infinitesimal interfacial width (sharp interface limit) evolving in physical phenomena can recover the actual interfacial dynamics. In this approach, the system of partial differential equations can be solved directly which allow the explicit implementation of boundary conditions at the interface.

Initially, the phase-field models were introduced by Fix [23] for solidification and other related areas to study interfacial dynamics. For example, dynamics which is given by a diffusion equation for the concentration in solidification problems or the temperature of

the bulk and the boundary conditions at the interface (conservation laws and equilibrium system) [24], having the sharp interface models. The phase-field model can be written by free energy functional having two parts, double well potential and gradient of phase-field:

$$E(u) = \int_{\Omega} (F(u) + \varepsilon^2 \nabla^2 u), \tag{1.1}$$

where $F(u)$ is double-well free energy functional which gives two minima as shown in figure 1.2. Different type of double well potential can be chosen, however, the most commonly use are in polynomial and logarithmic form [24]. In this work, we have used the double well potential as a polynomials of the forms $F(u) = u^6 - u^4 - u^2 + 1$ [25] and $F(u) = (1-u^2)^2$ [26].

These free energy functionals are formulated by considering some physical phenomena which contain parameter or family of parameters directly or indirectly related to width of interface. The parameters involved in the model would be chosen in such a way that interficial width approaches to zero, the diffuse interface model will approach to sharp interface model. Usually interfacial dynamics can be reconstructed by phase-field model where the thickness of interface is very small as compared to wide geometry of the problem (e.g., in solidification process, interfacial dynamics is capillary length that is on small scale). From numerical simulation perspective, such kind of models on small scale are intolerable. Karma and Rappel [13] have introduced the idea of thin

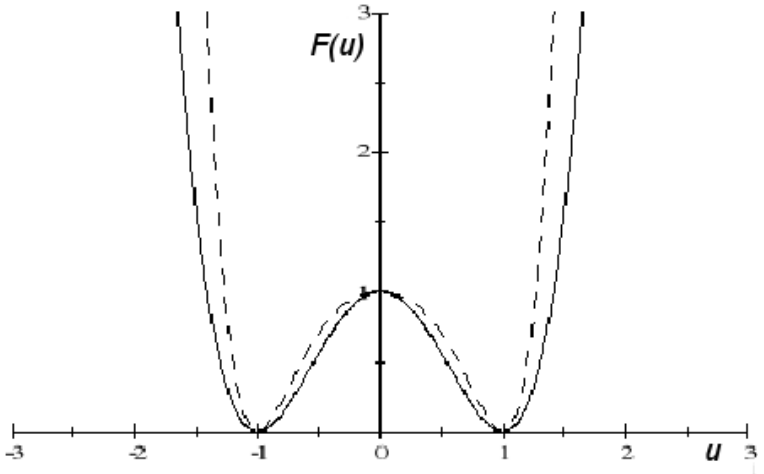


Figure 1.2: Double well potentials: $F(u) = u^6 - u^4 - u^2 + 1$ (dashed line) and $F(u) = (1-u^2)^2$ (solid line)

interface which allows to overlook small interface thickness. In the sharp-interface in two-dimensional case, the interface is mathematically like a surface with no structure or width. In the phase-field method, entire micro-structure is represented by continuous order parameter $u(\mathbf{x},t)$ such that:

$$u(\mathbf{x},t) = \begin{cases} 1, & \text{Phase-1} \\ -1, & \text{Phase-2.} \end{cases} \tag{1.2}$$

with a diffusion region of thickness ε, contains the interface $u(\mathbf{x},t) = 0$ as shown in figure 1.1(a). There are two kind of phase-field models based on Allen-Cahn and Cahn-Hilliard equations. Both equations evolving Ginzburg-Landau energy functional given in equation (1.1) with $F(u)$ a double well potential. Conventionally, the Allen-Cahn equation and Cahn-Hilliard equation are formulated in such a way that the energy for these models is gradient flow which means that the energy of the system is decreasing along the solution of the models. The Allen-Cahn equation has non-conserved dynamics while the Cahn-Hillard satisfy the the mass conservation law [13].

1.1.1 Allen-Cahn Equation

The Allen-Cahn equation which is a stiff nonlinear partial differential equation was first introduced by Allen and Cahn to describe the process of phase separation of a binary alloy [27]. The Allen-Cahn equation is non-conserved second order nonlinear partial differential equation that described the motion of anti-phase boundaries. Consider non-conserved variable as u is phase-field order parameter and split the Ginzburg-Landau energy functional as:

$$E = F(u) + G(\nabla u). \tag{1.3}$$

Total free energy functional E which measures the energy in the interface and elucidate the liquid and solid phases is given by.

$$E = \int_{\Omega} E\,(u, \nabla u)\, dV \tag{1.4}$$

To model the kinetics by minimizing of the free energy by

$$\delta E = \int_{\Omega} \left(\frac{\partial F}{\partial u}\delta u + \frac{\partial G}{\partial(\nabla u)}\delta \nabla u \right) dV. \tag{1.5}$$

Allen and Cahn [28] assumes that

$$\frac{\partial u}{\partial t} = -\delta E \tag{1.6}$$

From equations (1.5) and (1.6), we get

$$\frac{\partial u}{\partial t} = \varepsilon^2 \nabla^2 u - F'(u). \tag{1.7}$$

The no-flux boundary conditions $\nabla u \cdot \nu = 0$ on $\mathbf{x} \in \partial\Omega$ are assumed with ν is outward normal vector at the boundary domain.

1.1.2 Cahn-Hillard Equation

The Cahn-Hilliard equation was introduced in 1958 for the phase separation which happens in binary alloys [29]. The phase separation phenomenon can be seen in tumor tissues growth and polymers etc [30, 31]. It also plays crucial role where it has been used to formulate the nature of conserved order variables [32]. The chemical potential $c(\mathbf{x},t)$ which measures the energy of system that how much the energy changes with the change in configuration that is:

$$c(\mathbf{x},t) = \delta E = F^{'}(u) - \varepsilon^2 \nabla^2 u \tag{1.8}$$

By using the Fick's first law of diffusion, the continuity equation is given as follows [32]

$$\frac{\partial u}{\partial t} + \nabla \cdot (-M \nabla c(\mathbf{x},t)) = 0, \tag{1.9}$$

From equations (1.8) and (1.9), we get the following form of Cahn-Hilliard equation

$$\frac{\partial u}{\partial t} = \nabla.[M \nabla (F^{'}(u) - \varepsilon^2 \nabla^2 u)]. \tag{1.10}$$

where $M = M(u) = \frac{1}{4}(u^2-1)^2$ is the variable mobility which may be constant or usually depends on concentration $u(\mathbf{x},t)$. The equation (1.10) is a forth-order nonlinear PDE, can be split and rewritten as system of two second order PDEs as

$$\frac{\partial u}{\partial t} = \nabla.[M(u) \nabla c], \tag{1.11}$$

$$c(\mathbf{x},t) = F^{'}(u) - \varepsilon^2 \nabla^2 u. \tag{1.12}$$

The following boundary conditions $\nabla u \cdot \nu = \nabla c \cdot \nu = 0$ on $\mathbf{x} \in \partial\Omega$ are applied such that the total mass and energy are conserved [33].

1.2 Challenges in Simulation

In the last few decades, the scientific computing has emerged as a leading tool for studying the behavior of mathematical models representing some natural phenomena with due to advanced computer technology. Several numerical algorithms have been developed with wide range of applications in science and engineering. Numerical simulation of multi-phase flows problem is an interesting field due to development of more sophisticated phase-field model of interfacial dynamics. In recent years, different numerical techniques are competing for multi-phase flows as well as flow in complex geometries. Conventionally, Allen-Cahn and Cahn-Hilliard equations are stiff in nature and higher-order nonlinear parabolic PDEs. In the phase separation process, the simultaneous detected change leads to a sharp interface within the two phases but a very slow in the coarsening, which leads to bulk phase, wherein the interface separates them. Obviously, when the two sub-processes are described by different space and time scales, whose make this equation much difficult to solve analytically. There has been significant progress in the developing discretization techniques for solving Cahn-Hilliard and Allen-Cahn, both in terms of time and space discretization methods. Many time discretization schemes (e.g backward -Euler, semi-implicit and higher order implicit Runge-Kutta etc.) have been employed [34, 35]. Some adaptive time-stepping strategies are also proposed to reduce computational cost and attaining minimal error control [36, 37]. However, the multidisciplinary applications of the gradient flow equations requires development of time efficient numerical schemes, which is the aim of this work.

The two main components of this work are (i.) selection of high-order time discretization schemes and (ii.) appropriate adaptivity. Due to stiff nature of the governing equations, the explicit methods are not effective due to time-step restriction. Although the implicit schemes are an alternate approach with no restriction on the time-step at all [38, 39]. However, the nonlinear nature of these equations require solution of a nonlinear system of equations along with linear solver per time step after discretization. Many implicit schemes like Euler backward and Crank-Nicholson are utilized [40, 42, 35] for solving stiff and nonlinear equations. They allow adaptive time-stepping strategy to deal with the multi-scale phenomena, result in reducing computational costs. The implicit schemes are also unconditionally stable and allow for much larger time step. However,

the implicit time schemes are more reliable due to their unconditional stability and their flexibility to solve the Allen-Cahn/Cahn-Hilliard equations [41]. In implicit schemes, adaptive time-stepping can be easily applied with different strategies, e.g, by adapting the time-step size on the basis of required number of iterations [42], or changing the time-step size during the simulations [43].

Recently, some approaches base on implicit Runge-Kutta schemes are proposed [44, 45], particularly we are interested in implementation of diagonally implicit schemes for solving Allen-Cahn/Cahn-Hilliard equations. We developed the algorithm to solve Allen-Cahn/Cahn-Hilliard equations with focus on computational efficiency and robustness using DUNE-PDELab. The PDELab is a discretisation module based on Distributed and Unified Numerics Environment (DUNE), which is an open source software. The PDELab is rapid prototyping which substantially reduce time to implement discretisation and solver for system of PDEs. It is successfully used for numerical solution of many elliptic, parabolic and hyperbolic PDEs arising from different application areas.

1.3 Some Applications

1.3.1 Material Sciences

In material sciences, the properties of materials contain a link to their underlying microstructure. Most applications in civil engineering require high-strength steels that contain a mixture of fine crystal grains and dispersion of hard and soft phases in their microstructure. During recrystallization and solidification, pattern formation of first order of the crystal grains, a aggressive growth of these grain due to reproduction of the whole free energy of the system, but the kinetics limit the spread of heat and mass. Since the thermodynamic driving forces vary so, the reduction of free bulk-energy by controlled solidification and the anisotropy of surface energy as shown in figure 1.3. Due to development of new and advanced theoretical and numerical tools, it is recent explosion and very useful to model and predict materials properties and micro-structure on advanced level. Recently, due to advancement and new inventions in computer technology the modern parallel computing now simulates several billions atoms on the scale of nanoseconds. On the large scale levels different sharp and continuous interface methods have been developed which make it possible to quantitative modeling of free surface

kinetics involved in formation of the micro-structure. Usually, the the solidification process serves as sample of non-equilibrium phase transformations, which determine the formation of a micro-complex structure when materials are processed shown in figure 1.3.

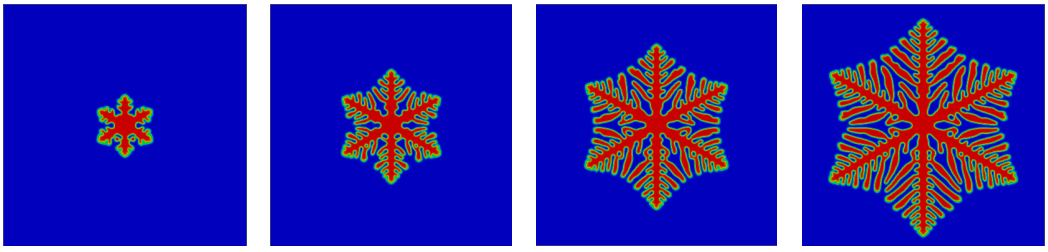


Figure 1.3: Dendritic crystal growth in materials.

Since it solidify at room temperature and transparent and gives a good look at process of solidification, due to this good properties this material is the favorite choice for among the researchers. Unexpectedly, the properties of the succinonitrile (SCN) remain unchanged qualitatively in metals and their alloys. The pattern formation of these materials is shown in figure 1.3 and is found in the solid state transformations.

1.3.2 Biological Sciences

In biology, the tumor growth is a complex phenomenon that take place at various scales but the cellular, molecular and tissue scales are more common. The well known example of most relevant phenomena that occur at the molecular level will include the absorption of nutrients DNA degradation, which leads to abnormal proliferation of tumor cells. Many cellular movement and activities such as apoptosis and necrosis occur at the cellular level [46]. The boundary between the cell and tissue scales is thin interface, but we consider the tissue scales to be effective when tumor angiogenesis can induce. Angiogenesis is the process of formation of new capillaries from blood vessels which already exist and it has important role for a tumor growth. Some types of cancer form solid tumors mutated from abnormal cells. These cells, which grow aggressively in the solid masses at the expense of all the resources of their environment. At this stage of tumor growth (avascular stage), cancer cells obtain the food due to diffusion process. However, when the tumor reaches a certain size, no more than 2 mm radius in experiments [47], it stops the overall growth. In avascular tumors growth, in absence of new blood vessels, the nutrient for the tumors by diffusion of adjacent cells, the tumor cells

consume more nutrients than healthy cells and supply do not go deep into the region. This results in an unequal delivery of nutrients to the tumor, mainly determined by the

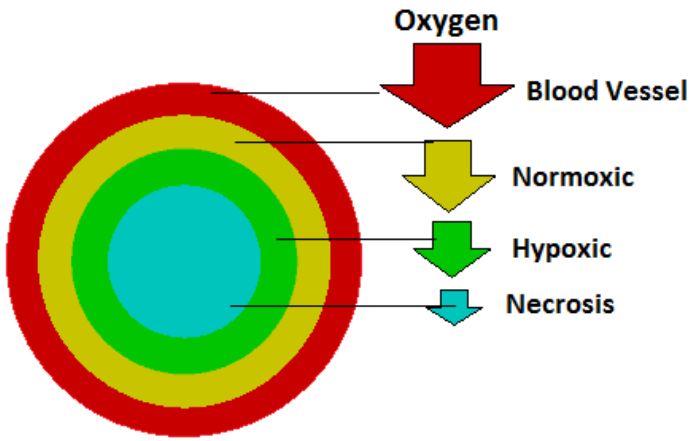


Figure 1.4: Different stages of tumor growth phenomena

distance from the surface of tumor. Based on the unequally distribution of nutrients, the tumor can be divided into three different regions, namely the necrotic nucleus, the hypoxic core and proliferative edge. The proliferating cells quickly share nutrients while in the necrotic nuclear nutrients are low, that the cells due to the lack of the blood and lymph vessels in the tumor necrosis remains debris as shown in figure 1.4. However, in hypoxic tumor cells are temporarily available in resting area and late apoptotic account without additional nutrients. The tumor radially divided into three parts, which is the proliferative (normoxic) edge (outer ring), the hypoxic zone (central part) and necrotic core (inner portion) [48, 49].

1.4 Publications

The present work comprises on following publications/submissions in well reputed international journals.

1. Abdullah Shah, Muhammad Sabir, Peter Bastain, An efficient time-stepping scheme for numerical simulation of dendritic crystal growth, European Journal of Computational Mechanics, 25(6), 475-488, 2017.

2. Muhammad Sabir, Abdullah Shah, Wazir Muhammad, Ijaz Ali, Peter Bastian, A mathematical model of tumor hypoxia targeting in cancer treatment and its

numerical simulation, Computers and Mathematics with Applications, 74(12)-3250-3259, 2017.

3. Abdullah Shah, Muhammad Sabir, Muhammad Qasim, Peter Bastain, An efficient numerical scheme for solution of Allen-Cahn equation, Numerical Methods for Partial Differential Equations (revision submitted).

4. Muhammad Sabir, Abdullah Shah, An adaptive time-stepping scheme for numerical solution of Cahn-Hilliard equation with variable mobility, Results in Physics, (under revision)

1.5 Summary

This chapter highlighted the main concepts related to this thesis and foundation for forthcoming chapters. Computational complexities and difficulties in solving phase-field models are discussed. Some possible applications of phase-field models are also given.

Chapter 2

Numerical Methods for Solving Phase-Field Models

Various numerical methods have been developed for numerical solution of nonlinear parabolic PDE. The most common are finite difference method (FDM), finite volume method (FVM) and finite element method etc. The FDM which is easy to implement for regular geometries, however, it needs some interpolation between grid points for irregular shapes domains with slope discontinuities, which may cause numerical oscillations. On the other hand finite element method (FEM) has geometrical benefits for irregular/complex boundaries. This chapter describes FEM based discretization technique for solving nonlinear parabolic PDEs with different time discretization schemes.

2.1 Numerical Discretization

This section provides implementation of spatial and temporal discretization schemes for parabolic PDEs. We start with nonlinear PDE and assuming $F'(u) = f(u)$ from equation 1.3 as:

$$\frac{\partial u}{\partial t} - \nabla^2 u + f(u) = 0 \qquad \text{in } \Omega \tag{2.1}$$

where $f(u)$ is a nonlinear source term or load function. The solution of equation (2.1) will also satisfy the boundary conditions on $\partial\Omega$ given by:

$$\hat{\alpha}u - \hat{\beta}\nabla u \cdot \nu = \begin{cases} g & \hat{\alpha} \neq 0,\ \hat{\beta} = 0 \\ j & \hat{\alpha} = 0,\ \hat{\beta} \neq 0 \end{cases} \tag{2.2}$$

$$u|_{t=o} = u_0 \tag{2.3}$$

where $\hat{\alpha}$ and $\hat{\beta}$ are some constants; and $\nabla u \cdot \nu$ is the directional derivative which is outer normal to the boundary. The problem (2.1) becomes boundary value problem by combining equations (2.1) and (2.2). If $\hat{\alpha} \neq 0$ and $\hat{\beta} = 0$ then the boundary value problem is referred to Dirichlet boundary value problem, also if $\hat{\alpha} = 0$ and $\hat{\beta} \neq 0$, then problem is referred to Neumann boundary value problem. On the other hand the third possibility is if both $\hat{\alpha}$ and $\hat{\beta}$ are non-zeros the problem goes to mixed boundary value problem, the very popular case for mixed type boundary conditions is Robin boundary conditions.

2.1.1 Space Discretization

For simplicity, we consider the solution method for steady state nonlinear Poisson equation of the form:

$$-\nabla^2 u + f(u) = 0 \quad \text{in } \Omega, \tag{2.4}$$

$$u = g \quad \text{on } \Gamma_D \subseteq \partial\Omega, \tag{2.5}$$

$$-\nabla u \cdot \nu = j \quad \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \tag{2.6}$$

$\Omega \subset \mathbb{R}^d$ is a domain, $f : \mathbb{R} \to \mathbb{R}$ is a given, possibly nonlinear function and ν denotes the unit outer normal to the domain. The weak formulation in the finite element method can be obtained by multiplying equation (2.4) with appropriate test function and using integration in space by parts to evaluate the integrals. The main advantage of FEM is to easily treat the full tensor and its flexibility in calculating mass and stiffness matrices analytically using global basis functions directly i.e.,

$$\text{Find } u \in U \text{ s.t.:} \quad r(u,v) = 0 \quad \forall v \in V, \tag{2.7}$$

with the continuous residual form

$$r(u,v) = \int_\Omega \nabla u \cdot \nabla v + (f(u))v\,dx + \int_{\Gamma_N} jv\,ds \tag{2.8}$$

and the function spaces $U = \{v \in H^1(\Omega) : \text{“}v = g\text{” on } \Gamma_D\}$ and $V = \{v \in H^1(\Omega) : \text{“}v = 0\text{” on } \Gamma_D\}$.

2.1.1.1 Finite Element Method

The finite element method [50, 51, 52, 53, 54, 55, 56] replaces the function spaces U and V by finite dimensional approximations defined on a finite element mesh. Before describing exactly how these spaces are constructed let us review some basic concepts.

2.1.1.2 Finite Element Space

For the detailed construction of the basis functions ϕ_j involves the finite element mesh, we define:

1. A set of vertices $\mathscr{X}_h = \{x_1, \ldots, x_N\}$ and elements $\mathscr{T}_h = \{T_1, \ldots, T_M\}$. Elements are closed and connected sets of points with non-intersecting interior partitioning the domain Ω.

2. A partitioning of the vertex index set $\mathscr{I}_h = \{1,\ldots,N\}$ into indices of interior and boundary vertices

$$\mathscr{I}_h = \mathscr{I}_h^{int} \cup \mathscr{I}_h^{\partial\Omega}, \quad \mathscr{I}_h^{int} = \{i \in \mathscr{I}_h : x_i \in \Omega\}, \quad \mathscr{I}_h^{\partial\Omega} = \{i \in \mathscr{I}_h : x_i \in \partial\Omega\}.$$

3. For every element $T \in \mathscr{T}_h$ a local-to-global map

$$g_T : \{0,\ldots,n_T-1\} \to \mathscr{I}_h$$

associating a local number of a corner of element T with a global vertex number. n_T is the number of corners of element T.

4. For every element $T \in \mathscr{T}_h$ an element transformation map

$$\mu_T : \hat{T} \to T \tag{2.9}$$

mapping the corresponding reference element to T. The element transformation map need not be affine but is assumed to be sufficiently differentiable with invertible Jacobian as well as consistent in the sense $\forall i \in \{0,\ldots,n_T-1\} : \mu_T(\hat{x}_i) = x_{g_T(i)}$.

The conforming finite element space of degree k in dimension d on the mesh $\mathscr{T}_h$ is given by

$$V_h^{k,d}(\mathscr{T}_h) = \left\{ v \in C^0(\overline{\Omega}) : \forall T \in \mathscr{T}_h : v|_T = \mu_T \circ p_T \wedge p_T \in \mathbb{P}_T^{k,d} \right\} \tag{2.10}$$

with the appropriate space of multivariate polynomials of degree k in dimension d depending on the type of element T:

$$\mathbb{P}_T^{k,d} = \begin{cases} \left\{ p : p(x_1,\ldots,x_d) = \sum\limits_{0 \le \|\alpha\|_1 \le k} c_\alpha x_1^{\alpha_1} \cdot \ldots \cdot x_d^{\alpha_d} \right\} & \hat{T} = \hat{S} \text{ (simplex)}, \\ \left\{ p : p(x_1,\ldots,x_d) = \sum\limits_{0 \le \|\alpha\|_\infty \le k} c_\alpha x_1^{\alpha_1} \cdot \ldots \cdot x_d^{\alpha_d} \right\} & \hat{T} = \hat{C} \text{ (cube)}. \end{cases} \tag{2.11}$$

Note that in dimension 1, there is no difference between cube and simplex. In dimension 2, triangular and quadrilateral elements may be mixed. In dimension 3, however, tetrahedral and hexahedral elements may not be mixed without introducing additional elements such as prisms. The dimension of $\mathbb{P}_T^{k,d}$ is

$$n_{\hat{C}}^{k,d} = (k+1)^d$$

in the case of a cube reference element and

$$n_{\hat{S}}^{k,d} = \begin{cases} 1 & k = 0 \vee d = 0 \\ \sum_{i=0}^{k} n_{\hat{S}}^{i,d-1} & \text{else} \end{cases}$$

in the case of a simplex reference element.

2.1.1.3 Local and Global Lagrange Basis

The equation (2.10) defines the finite element space without reference to a basis. For the implementation in computer, a basis is needed. We now generalize the construction of the Lagrange basis functions to the general space $V_h^{k,d}(\mathscr{T}_h)$. To that end, the reference element $\hat{T}$ is equipped with Lagrange points

$$L_{\hat{T}} = \left\{ \hat{x}_0^{\hat{T}}, \ldots, \hat{x}_{n_{\hat{T}}^{k,d}-1}^{\hat{T}} \right\},$$

and Lagrange polynomials

$$P_{\hat{T}} = \left\{ p_0^{\hat{T}}, \ldots, p_{n_{\hat{T}}^{k,d}-1}^{\hat{T}} \right\},$$

such that

$$p_i^{\hat{T}}(\hat{x}_j^{\hat{T}}) = \delta_{i,j}.$$

Global continuity is then ensured by carefully placing $n_{\hat{T}}^{k,d-c_0}$ of these points on each sub-entity of co-dimension c_0 in the reference element. The local-to-global map g_T is extended to map indices of local basis functions to indices of global basis functions of the finite element space. Let

$$g_T : \{0, \ldots, n_{\hat{T}}^{k,d} - 1\} \to \mathscr{I}\left(V_h^{k,d}(\mathscr{T}_h)\right) = \left\{0, \ldots, \dim V_h^{k,d}(\mathscr{T}_h) - 1\right\},$$

be this extension and

$$C(i) = \{(T,m) \in \mathscr{T}_h \times \mathbb{N} : g_T(m) = i\}$$

its inversion. Then

1. $C(i)$ is nonempty for all $i \in \mathscr{I}\left(V_h^{k,d}(\mathscr{T}_h)\right)$

   and

2. for all $(T,m),(T',m') \in C(i)$ it holds $\mu_T(x_m^{\hat{T}}) = \mu_{T'}(x_{m'}^{\hat{T}'})$.

The global Lagrange basis functions spanning $V_h^{k,d}(\mathscr{T}_h)$ are then defined as

$$\phi_i(x) = \begin{cases} p_m^{\hat{T}}(\mu_T^{-1}(x)) & x \in T \wedge (T,m) \in C(i) \\ 0 & \text{else} \end{cases}, \quad i \in \mathscr{I}\left(V_h^{k,d}(\mathscr{T}_h)\right).$$

Each global basis function ϕ_i corresponds to a Lagrange point x_i from the ordered set

$$\mathscr{X}_h^{k,d} = \left\{x_i \in \overline{\Omega} : x_i = \mu_T(\hat{x}_m^{\hat{T}}) \wedge (T,m) \in C(i)\right\}.$$

Note that for degree $k = 1$ one has $\mathscr{I}\left(V_h^{1,d}(\mathscr{T}_h)\right) = \mathscr{I}_h$, i.e. one basis function is associated with each vertex of the mesh.

2.1.1.4 Incorporation of Dirichlet Boundary Conditions

The standard way to handle Dirichlet boundary conditions in the conforming finite element method is to incorporate them directly into the function space. To do that define the indices of Lagrange points on the Dirichlet boundary

$$\mathscr{I}^D\left(V_h^{k,d}(\mathscr{T}_h)\right) = \left\{i \in \mathscr{I}\left(V_h^{k,d}(\mathscr{T}_h)\right) : x_i \in \mathscr{X}_h^{k,d} \wedge x_i \in \Gamma_D\right\}.$$

For the test space one defines the space of finite element functions that are zero on the Dirichlet boundary:

$$V_{h,0}^{k,d}(\mathscr{T}_h) = \left\{v \in V_h^{k,d}(\mathscr{T}_h) : v(x_i) = 0 \quad \forall i \in \mathscr{I}^D\left(V_h^{k,d}(\mathscr{T}_h)\right)\right\}.$$

For the trial space set

$$u_{h,g} = \sum_{i \in \mathscr{I}^D\left(V_h^{k,d}(\mathscr{T}_h)\right)} g(x_i)\phi_i,$$

with the Dirichlet boundary condition function g. In fact one may want to use

$$u_{h,g} = \sum_{i \in \mathscr{I}\left(V_h^{k,d}(\mathscr{T}_h)\right)} u_g(x_i)\phi_i \qquad \text{with } u_g(x_i) = g(x_i) \text{ for all } i \in \mathscr{I}^D\left(V_h^{k,d}(\mathscr{T}_h)\right)$$

, which incorporates the Dirichlet boundary conditions on Γ_D and provides an initial guess for the nonlinear iterative solver in the interior of the domain. Then the trial space is

$$U_h^{k,d}(\mathscr{T}_h) = \left\{u \in V_h^{k,d}(\mathscr{T}_h) : u = u_{h,g} + w \wedge w \in V_{h,0}^{k,d}(\mathscr{T}_h)\right\}.$$

Finally, the finite element problem in its precise form reads

$$\text{Find } u \in U_h^{k,d}(\mathscr{T}_h) \text{ s.t.:} \quad r(u,v) = 0 \quad \forall v \in V_{h,0}^{k,d}(\mathscr{T}_h). \tag{2.12}$$

Given a finite-dimensional space $U_h = \text{span}\left\{\phi_j : j \in J_h = \{1,\ldots,n\}\right\}$ a subspace $\tilde{U}_h$ is constructed by

1. selecting a subset of indices $\tilde{J}_h \subset J_h$,

2. and setting $\tilde{U}_h = \text{span}\left\{\tilde{\phi}_j : j \in \tilde{J}_h\right\}$, where the new basis functions have the form

$$\tilde{\phi}_j = \phi_j + \sum_{l \in J_h \setminus \tilde{J}_h} (B)_{j,l} \phi_l \quad \forall j \in \tilde{J}_h.$$

Thus, any subspace of U_h is characterized by $C = (\tilde{J}_h, B)$. This abstractions allows to represent Dirichlet conditions ($J_h \setminus \tilde{J}_h$ are the indices of the Dirichlet nodes and $B = 0$), hanging nodes ($J_h \setminus \tilde{J}_h$ are the indices of hanging nodes and B represents the interpolation conditions) or even rigid body modes.

2.1.1.5 Element-wise Computations

We now turn to how the residual can be evaluated in practice. The residual form ((2.8)) can be readily decomposed into element-wise contributions:

$$r(u,v) = \sum_{T \in \mathscr{T}_h} \int_T \nabla u \cdot \nabla v + f(u) v \, dx, + \sum_{F \in \mathscr{F}_h^{\partial\Omega}} \int_{F \cap \Gamma_N} jv \, ds. \tag{2.13}$$

Here $\mathscr{F}_h^{\partial\Omega}$ is the set of intersections of elements with the domain boundary, i.e. $F = T_F^- \cap \partial\Omega$ where T_F^- is the element associated with F. The three terms in equation (2.13) can now be evaluated by extension in maps μ_T form (2.9) we have

$$\nabla u(\mu_T(\hat{x})) = J_{\mu_T}^{-1}(\hat{x}) \hat{\nabla} \hat{u}(\hat{x})$$

, with $J_{\mu_T}(\hat{x})$ the Jacobian of μ_T at point $\hat{x}$.

2.1.1.6 Boundary Terms

For $F \in \mathscr{F}_h^{\partial\Omega}$ with $F \cap \Gamma_N \neq \emptyset$ and $(T_F^-, m) \in C(i)$, we obtain

$$\int_F jv \, ds = \int_{\hat{F}} j(\mu_F(\hat{x})) p_m^{\hat{T}}(\eta_F(\hat{x})) \sqrt{|\det(J_{\mu_F}^T(\hat{x}) J_{\mu_F}(\hat{x}))|} \, ds,$$

The map μ_F maps the reference element $\hat{F}$ of F into global coordinates while the map η_F maps $\hat{F}$ into the reference element $\hat{T}$ of T. Also the integration element has to be redefined accordingly. Again, all contributions of the face F can be collected in a vector:

$$\int_F jv \, ds = \int_{\hat{F}} j(\mu_F(\hat{x})) p_m^{\hat{T}}(\eta_F(\hat{x})) \sqrt{|\det J_{\mu_T}^T(\hat{x}) J_{\mu_T}(\hat{x})|} \, ds. \tag{2.14}$$

2.1.1.7 Diffusion and Nonlinear Terms

For any $(T,m) \in C(i)$, we get

$$\begin{aligned}\int_T \nabla u \cdot \nabla \phi_i + f(u)\phi_i \, dx, &= \int_T \sum_j (z)_j \left(\nabla \phi_j \cdot \nabla \phi_i\right) + f\left(\sum_j (z)_j \phi_j\right) \phi_i \, dx, \\ &= \int_{\hat{T}} \sum_n (z)_{g_T(n)} (J_{\mu_T}^{-1}(\hat{x}) \hat{\nabla} p_n^{\hat{T}}(\hat{x})) \cdot (J_{\mu_T}^{-1}(\hat{x}) \hat{\nabla} p_m^{\hat{T}}(\hat{x})) \\ &\quad + f\left(\sum_n (z)_{g_T(n)} p_n^{\hat{T}}(\hat{x})\right) p_m^{\hat{T}}(\hat{x}) |\mathrm{det} J_{\mu_T}(\hat{x})| \, dx.\end{aligned}$$

Again contributions for all test functions can be collected in a vector

$$\begin{aligned}(\mathscr{R}_T^V(R_T z))_m = \sum_n (z)_{g_T(n)} &\int_{\hat{T}} (J_{\mu_T}^{-1}(\hat{x}) \hat{\nabla} p_n^{\hat{T}}(\hat{x})) \cdot (J_{\mu_T}^{-1}(\hat{x}) \hat{\nabla} p_m^{\hat{T}}(\hat{x})) |\mathrm{det} J_{\mu_T}(\hat{x})| \, dx \\ &+ \int_{\hat{T}} f\left(\sum_n (z)_{g_T(n)} p_n^{\hat{T}}(\hat{x})\right) p_m^{\hat{T}}(\hat{x}) |\mathrm{det} J_{\mu_T}(\hat{x})| \, dx.\end{aligned} \tag{2.15}$$

Now from $(2.14 - 2.15)$, evaluation of the algebraic residual is

$$R(z) = \sum_{T \in \mathscr{T}_h} R_T^T \mathscr{R}_T^V(R_T z) + \sum_{T \in \mathscr{T}_h} R_T^T \mathscr{L}_T^V + \sum_{F \in \mathscr{F}_h^{\partial\Omega} \cap \Gamma_N} R_T^T \mathscr{L}_F^B. \tag{2.16}$$

The Jacobian of the residual is

$$(J(z))_{i,j} = \frac{\partial R_i}{\partial z_j}(z) = \sum_{(T,m,n):(T,m)\in C(i) \wedge (T,n) \in C(j)} \frac{\partial (\mathscr{R}_T^V)_m}{\partial z_n}(R_T z).$$

2.1.1.8 Algebraic Problem

Any finite-dimensional function space is spanned by a basis. So, assume that

$$U_h = \mathrm{span}\{\phi_1, \ldots, \phi_n\}, \quad V_h = \mathrm{span}\{\psi_1, \ldots, \psi_m\},$$

are corresponding sets of basis functions for U_h and V_h. Expanding the solution $u_h = \sum_{j=1}^n (z)_j \phi_j$ in the basis and hereby introducing the coefficient vector $z \in \mathbb{R}^n$, we can reformulate the problem as

$$\begin{aligned}&\text{Find } u_h \in U_h \text{ s.t.:} && r(u_h, v) = 0 && \forall v \in V_h \\ &\Leftrightarrow && r\left(\sum_{j=1}^n (z)_j \phi_j, \psi_i\right) = 0 && \forall i = 1, \ldots, m \\ &\Leftrightarrow && R(z) = 0, &&\end{aligned}$$

where $R : \mathbb{R}^n \to \mathbb{R}^m$ given by $R_i(z) = r_h\left(\sum_{j=1}^n (z)_j \phi_j, \psi_i\right)$ is a nonlinear, vector-valued function. The solution of the nonlinear algebraic equation $R(z) = 0$, is typically computed in an iterative fashion using e.g. a fixed-point iteration of the form

$$z^{(k+1)} = G(z^{(k)}) = z^{(k)} - \lambda^k W(z^{(k)}) R(z^{(k)}). \tag{2.17}$$

Here λ^k is a damping factor and $W(z^{(k)})$ is a preconditioner matrix, e.g. in Newton's method (see e.g. [54]) one has

$$W(z^{(k)}) = (J(z^{(k)}))^{-1} \quad \text{where } (J(z^{(k)}))_{i,j} = \frac{\partial R_i}{\partial z_j}(z^{(k)}).$$

We now assumed that $n = m$ and that the Jacobian $J(z^{(k)})$ is invertible. Newton's method requires the solution of the linear system $J(z^{(k)})w = R(z^{(k)})$ in each step which could be done using either direct or iterative methods. The implementation of Newton's method requires the following algorithmic building blocks:

1. residual evaluation $R(z)$,
2. Jacobian evaluation $J(z)$,
3. matrix-free Jacobian application $J(z)w$.

Only one of the methods 2. and 3. is required depending on the chosen solution procedure.

2.1.1.9 Matrix-free Evaluation

The matrix-free multiplication of the Jacobian $J(z)$ with a vector w is, with the definitions above:

$$(J(z)w)_i = \sum_{j=1}^n (J(z))_{i,j}(w)_j = \sum_{j=1}^n \frac{\partial}{\partial z_j} r_h \left(\sum_{l=1}^n (z)_l \phi_l, \psi_i \right) (w)_j.$$

At this point one may exploit the local support of basis functions in order to compute only the partial derivatives that are nonzero.

In the linear case, for comparison, one has $r_h(u,v) = a_h(u,v) - l_h(v)$ where a_h is a bilinear form and l_h is a linear form. Then, the application of the Jacobian can be

simplified as

$$
\begin{aligned}
(J(z)w)_i &= \sum_{j=1}^{n} \frac{\partial}{\partial z_j} r_h \left(\sum_{l=1}^{n} (z)_l \phi_l, \psi_i \right) (w)_j \\
&= \sum_{j=1}^{n} \frac{\partial}{\partial z_j} \left(a_h \left(\sum_{l=1}^{n} (z)_l \phi_l, \psi_i \right) - l_h(\psi_i) \right) (w)_j \\
&= \sum_{j=1}^{n} \frac{\partial}{\partial z_j} \left(\sum_{l=1}^{n} (z)_l a_h(\phi_l, \psi_i) \right) (w)_j \\
&= \sum_{j=1}^{n} a_h(\phi_j, \psi_i)(w)_j = a_h \left(\sum_{j=1}^{n} (w)_j \phi_j, \psi_i \right) = (Aw)_i,
\end{aligned}
$$

where $(A)_{i,j} = a_h(\phi_j, \psi_i)$ is the stiffness matrix which is independent of z. For this reason, there exist two different functions for matrix-free operator application, one for the linear case providing only the argument w and one for the nonlinear case providing two arguments z and w. Note also that it is advantageous to separate in the implementation of the residual form the part that depends on trial and test functions, and which consequently contributes to the Jacobian, and the part that only depends on the test functions and which does not contribute to the Jacobian.

2.1.1.10 Adaptive Grid Refinement

In this subsection, we provide the solution method for nonlinear Poisson equation using adaptive grid refinement. The finite element solution on a given mesh is used to compute local error indicators that can be used to iteratively reduce the discretization error. It is assumed that the nonlinearity of the PDE is not too strong, so that it may be approximated by linearizing around the current finite element solution. This allows the calculation of local error estimates that quantify the discretization error on each element of the mesh. In the previous sub-subsections, a fixed finite element space V_h was used, based on a fixed finite element mesh with an ordered set

$$\mathscr{X}_h = \{x_1, \ldots, x_N\}, \tag{2.18}$$

of vertices and an ordered set

$$\mathscr{T}_h = \{T_1, \ldots, T_M\}, \tag{2.19}$$

of elements. The task of choosing an appropriate mesh which is dependent on geometry of the problem is discussed in next chapters.

Any choice of finite element space will lead to discretization errors due to the finite-dimensional approximation of the true solution of the PDE, but the size of this error can be controlled by choosing an adequate mesh. Let $u_h \in V_h$ be the finite element solution and $\|\cdot\|$ an appropriate norm. The goal is then

$$\|u - u_h\| \leq \text{TOL}$$

with a given tolerance TOL, while keeping the complexity and size of V_h as low as possible.

The *a-priori* error estimates of formal proofs are not suitable for this task, since they contain a constant C that is typically unknown and don't provide information about the spatial distribution of the error. For this reason one is interested in *a-posteriori* error estimates of the form

$$\|u - u_h\| \leq \gamma(u_h) = \left(\sum_{T \in \mathscr{T}_h} \gamma_T^2(u_h) \right)^{1/2}, \tag{2.20}$$

where $\gamma_T(u_h)$ is a *local error estimator* for the element T. Such an error estimate should:

- have comparatively low cost of calculation to make an adaptive approach feasible
- accurately describe the spatial distribution and size of the discretization error

The local estimates $\gamma_T(u_h)$ can then be used in a strategy that tries to minimize $\gamma(u_h)$ for fixed $\dim(V_h)$ and thereby find a mesh with low discretization error $\|u - u_h\|$.

If the considered PDE is linear, i.e. $f(u)$ is an affine linear function, then it can be shown [52, 53] that such an estimate of the discretization error is given by

$$\gamma_T^2(u_h) = h_T^2 \|R_T(u_h)\|_{0,T}^2 + \sum_{F \in \mathscr{F}_h \cap \partial T} h_T \|R_F(u_h)\|_{0,F}^2,$$

with the *element residuals*

$$R_T(u_h) = \nabla^2 u_h + f(u_h),$$

and the *face residuals*

$$R_F(u_h) = \begin{cases} 2^{-1/2}[-(\nabla u_h) \cdot \nu] & F \in \mathscr{F}_h^i \\ -(\nabla u_h) \cdot \nu - j & F \in \mathscr{F}_h^N. \end{cases}$$

Here h_T is the local mesh width of the element T, $\mathscr{F}_h$ the set of faces of the elements in $\mathscr{T}_h$, ∂T the boundary of element T, $\mathscr{F}_h^i \subset \mathscr{F}_h$ the set of interior faces, $\mathscr{F}_h^N \subset \mathscr{F}_h$

the set of Neumann boundary faces and ν the unit normal vector of the face F pointing from one of the elements T_F^- that belong to F to the other T_F^+. $\|\cdot\|_{0,T}$ is the L^2 norm on T, $\|\cdot\|_{0,F}$ that on F, and $[\cdot]$ the jump of the given expression across the face F in the direction of ν. Broadly speaking, the element residual R_T measures to which degree u_h solves the PDE on the element T, and the face residual R_F measures the consistency of the flux $-(\nabla u_h)\cdot\nu$ between the elements.
With these definitions one can show that

$$\|u-u_h\|_{1,\Omega} \leq C\gamma(u_h) = C\left(\sum_{T\in\mathscr{T}_h}\gamma_T^2(u_h)\right)^{1/2},$$

with $\|\cdot\|_{1,\Omega}$ the H^1 norm on Ω. The right hand side of this inequality does not depend on u, which means it can be evaluated and used to control the discretization error.
Several aspects have to be considered:

- The derivation of the error estimator given above does not require additional regularity beyond $u \in H^1(\Omega)$. This is critical, since error control is especially important in problems with low regularity.

- The constant C in the estimate is usually not known exactly, and this should be taken into account when defining the tolerance TOL.

- The error estimate may converge with a lower order than the actual discretization error which yields a very pessimistic stopping criterion. If the error estimate converges with the same order, then the estimator is called *efficient*. The lowest applicable constant C is then the *efficiency index*.

All of these considerations only hold if PDE is linear. If the function $f(u)$ is nonlinear, then the estimate $\gamma(u_h)$ is no longer guaranteed to be an upper bound for the discretization error $\|u-u_h\|_{1,\Omega}$. However, it may still be used as an error indicator that is used to refine a given mesh. This can produce misleading results, and therefore the conclusions drawn from the estimate should be rather conservative. The error estimate will become more reliable the closer the finite element solution u_h is to the exact solution u, as long as the function $f(u)$ is regular enough to be linearized around the exact solution.

2.1.2 Temporal Discretization

We consider the following problem:

$$\begin{aligned} \partial_t u - \nabla^2 u + f(u) &= 0 && \text{in } \Omega \times \Sigma, \\ u &= g && \text{on } \Gamma_D \subseteq \partial\Omega, \\ -\nabla u \cdot \nu &= j && \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D, \\ u &= u_0 && \text{at } t = 0. \end{aligned}$$

The parameter functions g, j may now also depend on time and (with some restrictions) the subdivision into Dirichlet and Neumann boundary can be time-dependent as well. The initial condition u_0 is a function of $x \in \Omega$. Multiplying with a test function and integrating in space results in the following weak formulation [55]: Find $u \in L_2(t_0, t_0 + T; u_g + V(t))$:

$$\frac{d}{dt}\int_\Omega uv\,dx + \int_\Omega \nabla u \cdot \nabla v + f(u)v\,dx + \int_{\Gamma_N} jv\,ds = 0 \qquad \begin{array}{l} \forall v \in V(t), \\ t \in \Sigma, \end{array} \tag{2.21}$$

where $V(t) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni u_g(t)|_{\Gamma_D} = g$. This can be written in a more compact way with residual forms as follows:

$$\frac{d}{dt} m(u, v) + r(u, v) = 0 \quad \forall v \in V(t), \quad t \in \Sigma.$$

where

$$m(u, v) = \int_\Omega uv\,dx,$$

is the new temporal residual form, actually the L_2 inner product, and

$$r(u, v) = \int_\Omega \nabla u \cdot \nabla v + (f(u))v\,dx + \int_{\Gamma_N} jv\,ds,$$

is the spatial residual form known from equation (2.8). Under suitable assumptions it can be shown that problem (2.21) has a unique solution, see [55] for the linear case.

2.1.3 Discrete Form

In order to arrive at a fully discrete formulation, we follow the method of lines paradigm:

1. Choose a finite-dimensional test space $V_h(t) \subset V(t)$. Then (2.21) results in a system of ordinary differential equations for the coefficients $z_j(t)$ in the expansion of $u_h(t) = \sum_{j=1}^n (z_j(t))_j \phi_j$.

2. Choose an appropriate method to integrate the system of ordinary differential equations (ODEs).

For the finite-dimensional space we choose here is just the conforming finite element space $V_h^{k,d}(\mathscr{T}_h,t)$. It may now depend also on time due to the time-dependent splitting of the boundary in Dirichlet and Neumann part. Also the function $u_{h,g}(t)$ depends on time and we have $u_h(t) \in U_h(t) = u_{h,g}(t) + V_h(t)$.

For the integration of the system of ODEs, subdivide the time interval into not necessarily equidistant subintervals:

$$\overline{\Sigma} = \{t^0\} \cup (t^0, t^1] \cup \ldots \cup (t^{N-1}, t^N],$$

with $t^0 = t_0$, $t^N = t_0 + T$, $t^{k-1} < t^k$ for $1 \leq k \leq N$ and set the time-step to $\Delta t^k = t^{k+1} - t^k$. One of the simplest ODE integrators is the one-step-θ scheme that reads; Find $u_h^{k+1} \in U_h(t^{k+1})$ such that

$$\frac{1}{\Delta t^k}(m_h(u_h^{k+1}, v; t^{k+1}) - m_h(u_h^k, v; t^k)) + \theta r_h(u_h^{k+1}, v; t^{k+1}) + (1-\theta) r_h(u_h^k, v; t^k) = 0 \qquad \forall v \in V_h(t^{k+1}).$$

Reordering terms for $(k+1)$ and k show that this method results in the solution of a nonlinear system per time step which has the same structure as before [61], i.e., find $u_h^{k+1} \in U_h(t^{k+1})$ such that

$$m_h(u, v; t^{k+1}) + \Delta t^k \theta r_h(u, v; t^{k+1}) - m_h(u_h^k, v; t^k) + \Delta t^k (1-\theta) r_h(u_h^k, v; t^k) = 0 \qquad \forall v \in V_h(t^{k+1})$$

Implementation-wise the new residual form is comprised of a linear combination of temporal and spatial residual forms. For the temporal and spatial residual forms m_h and r_h, a class of methods can be written in the Shu-Osher form [57]:

1. $u_h^{(0)} = u_h^k$.

2. For $i = 1, \ldots, \bar{s} \in \mathbb{N}$, find $u_h^{(i)} \in u_{h,g}(t^k + d_i \Delta t^k) + V_h(t^{k+1})$:

$$\sum_{j=0}^{s} \Bigg[a_{ij} m_h \left(u_h^{(j)}, v; t^k + d_j \Delta t^k \right) + b_{ij} \Delta t^k r_h \left(u_h^{(j)}, v; t^k + d_j \Delta t^k \right) \Bigg] = 0, \qquad \forall v \in V_h(t^{k+1}).$$

3. $u_h^{k+1} = u_h^{(\bar{s})}$.

Here, we assume that the same type of boundary condition holds through the interval $(t^k, t^{k+1}]$. The parameter $\bar{s}$ denotes the number of stages of the scheme. An $\bar{s}$-stage scheme is given by the parameters

$$A = \begin{bmatrix} a_{10} & \dots & a_{1\bar{s}} \\ \vdots & & \vdots \\ a_{\bar{s}0} & \dots & a_{\bar{s}\bar{s}} \end{bmatrix}, \quad B = \begin{bmatrix} b_{10} & \dots & b_{1\bar{s}} \\ \vdots & & \vdots \\ b_{\bar{s}0} & \dots & b_{\bar{s}\bar{s}} \end{bmatrix}, \quad d = \begin{bmatrix} d_0 \\ \vdots \\ d_{\bar{s}} \end{bmatrix}.$$

Explicit schemes are characterized by $a_{ij} = 0$ for $j > i$ and $b_{ij} = 0$ for $j \geq i$. *Diagonally implicit* schemes are characterized by $a_{ij} = b_{ij} = 0$ for $j > i$. Some explicit and implicit methods can be brought to the Shu-Osher form as follows:

- One step θ scheme [58]:

$$A = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1-\theta & \theta \end{bmatrix}, \quad d = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

  Explicit/implicit Euler ($\theta \in \{0, 1\}$), Crank-Nicolson ($\theta = \frac{1}{2}$).

- Heun's second order explicit method [59]:

$$A = \begin{bmatrix} -1 & 1 & 0 \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix}, \quad d = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

- Shu's third order explicit scheme [60]:

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ -\frac{3}{4} & -\frac{1}{2} & 1 & 0 \\ -\frac{1}{3} & 0 & -\frac{2}{3} & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{2}{3} & 0 \end{bmatrix}, \; d = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{2} \\ 1 \end{bmatrix}.$$

- Alexander's two-stage second order strongly S-stable method [61]:

$$A = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & \theta & 0 \\ 0 & 1-\theta & \theta \end{bmatrix}, \quad d = \begin{bmatrix} 0 \\ \theta \\ 1 \end{bmatrix},$$

  with $\theta = 1 - \frac{\sqrt{2}}{2}$.

- Alexander's three-stage third order strongly S-stable method [61]:

$$A=\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \quad B=\begin{bmatrix} 0 & 2\theta & 0 & 0 \\ 0 & \theta'-\theta & \theta & 0 \\ 0 & \theta'' & \theta''' & \theta \end{bmatrix}, \quad d=\begin{bmatrix} 0 \\ \theta \\ \theta' \\ 1 \end{bmatrix},$$

with $\theta = 1-\frac{\sqrt{2}}{2}$, $\theta' = \frac{1+\theta}{2}$, $\theta'' = \frac{-1}{4}(6\theta^2-16\theta+1)$ and $\theta''' = \frac{1}{4}(6\theta^2-20\theta+5)$.

- Fractional step θ scheme, a three stage second order strongly A-stable scheme [62]:

$$A=\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \quad B=\begin{bmatrix} \theta\theta' & 2\theta^2 & 0 & 0 \\ 0 & 2\theta\theta' & 2\theta^2 & 0 \\ 0 & 0 & \theta\theta' & 2\theta^2 \end{bmatrix}, \quad d=\begin{bmatrix} 0 \\ \theta \\ 1-\theta \\ 1 \end{bmatrix},$$

with $\theta = 1-\frac{\sqrt{2}}{2}$, $\theta' = 1-2\theta$.

At each time level the algebraic system is solved by using BiConjugate Gradient (BiCG) method [56] and SuperLU [63]. For numerical simulation, we have used the PDELab [64, 65, 66, 67, 68] open source software which based on Distributed and Unified Numerics Environment (DUNE) [63].

2.2 Summary

In this chapter, we have provided the formulation and implementation of space-time discretization schemes in detail. For space discretization, the Galerkin FEM with uniform and adaptive grids, and a number of time discretization schemes have been discussed in detail. This chapter also provides a window for numerical implementation of the proposed schemes in forthcoming chapters for solving some challenging problems.

Chapter 3

Numerical Solution of Allen-Cahn Equation

This chapter presents an efficient algorithm for numerical solution of Allen-Cahn equation representing the motion of anti-phase boundaries. The proposed algorithm is based on the diagonally implicit fractional step θ-scheme for time discretization and conforming FEM for space discretization. For steady state solution, we use both uniform and adaptive grids and illustrate the effectiveness of adaptive grids over uniform grids by estimating the error norm. For unsteady case, the robustness and accuracy of the proposed diagonally implicit fractional step θ-scheme is compared with existing schemes (e.g. explicit, diagonally implicit and implicit) in term of computational cost and temporal error. The algorithm is also applied to simulate some test problems.

3.1 Introduction

The problem of interfacial dynamics in multi-phase flows is multi-disciplinary with several industrial applications. The Allen-Cahn equation (phase-field model) [27] is gaining popularity in research community for tracking the interface in multi-phase flows. Numerical investigation of such problems are challenging due to jump discontinuity in the material properties at interface. Therefore, robust and adaptive numerical techniques are required to simulate such problems. However, numerical methods for solving such type of stiff PDEs require more work per time-step. Explicit methods are conditionally stable for such problems but implicit methods are more numerically stable even with larger time-step size. Also in multi-phase flow problems, the width of transition regions (interface) is very thin which required clustering of grid points near the interface to accurately resolve the interfacial region. Therefore, for space discretization several methods are proposed to reduce the computational cost of simulation by using the adaptive grids refinement near the interface [69]. For temporal discretization, Alexander has used some diagonally implicit Runge-Kutta (DIRK) time discretization methods with different order of accuracy for solving stiff PDEs [61].

In this chapter, we have used the P_1 conforming FEM [70, 71, 72] and diagonal implicit fractional step θ-scheme (DIFST) [62] for spatial and temporal discretization respectivelyto solve the Allen-Cahn equation numerically. Some other temporal discretization schemes (e.g implicit Euler [58], Heun's [59], Shu's [60] and Alexander's [61]) are also used for comparison purpose.

3.2 Numerical Discretization

We begin by considering a non-conserved phase-field variable u, governed by the Allen-Cahn equation [69]:

$$\begin{aligned} \frac{\partial u}{\partial t} - \varepsilon \nabla^2 u + f(u) = 0, \qquad & \text{in } \Omega, \\ u = g, \qquad & \text{on } \Gamma_D \subseteq \partial\Omega, \\ -\nabla u \cdot \nu = 0, \qquad & \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \end{aligned} \tag{3.1}$$

Here $\Omega \subset \mathbb{R}^d$, $f : \mathbb{R} \to \mathbb{R}$ is a nonlinear function given by $f(u) = 6u^5 - 4u^3 - 2u$, ε is interfacial width and ν denotes the unit outer normal.

3.2.1 Space Discretization

For space discretization, we consider the steady state form of of equation (3.1) as:

$$\begin{aligned} -\varepsilon \nabla^2 u + f(u) = 0, \qquad & \text{in } \Omega, \\ u = g, \qquad & \text{on } \Gamma_D \subseteq \partial\Omega, \\ -\nabla u \cdot \nu = 0, \qquad & \text{on } \Gamma_N = \partial\Omega \setminus \Gamma_D. \end{aligned} \tag{3.2}$$

The weak formulation of equation (3.2) is:

$$\text{find } u \in U \text{ such that} \quad r(u,v) = 0 \quad \forall v \in V, \tag{3.3}$$

with the continuous residual form

$$r(u,v) = \int_\Omega \varepsilon \nabla u \cdot \nabla v + (f(u))v\,dx = a(u,v) + l(u), \tag{3.4}$$

where $a(u,v) = \int_\Omega \varepsilon \nabla u \cdot \nabla v$ and $l(u) = \int_\Omega (f(u))v$. In equation (3.3), $U = \{v \in H^1(\Omega) : v = g \text{ on } \Gamma_D\}$ and $V = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}$ are trial and test function spaces respectively. The finite element solution on a given mesh is used to compute the local error indicator that can reduce the discretization error iteratively.

Let $u_h \in V_h$ be the finite element solution with an appropriate norm and we are compute *a-posteriori* error estimate while keeping the complexity and size of V_h as low as possible. A simple residual-based a posterior estimate for equation (3.2) would be:

$$\|u - u_h\|_{L_2} \leq ch \| -\varepsilon \nabla^2 u_h + f(u_h)\|_{H_0^1} = \gamma(u_h) \leq \left(\sum_{T \in \mathcal{T}_h} \gamma_T^2(u_h) \right)^{1/2}, \tag{3.5}$$

where $\gamma_T(u_h)$ is a local error estimator for the element T and $\mathscr{T}_h = \{T_1, \ldots, T_M\}$ is the set of elements, and H_0^1 is Hilbert space. In a posteriori estimates, the right-hand side depends on h and u_h, but not on u which could be evaluated after computing u_h. The error estimates have the following properties:

- have comparatively low cost of calculations to make an adaptive approach feasible,
- accurately describe the spatial distribution and size of the discretization error.

The local estimates $\gamma_T(u_h)$ can be used as a strategy to minimize $\gamma(u_h)$ for fixed $\dim(V_h)$ and thereby find a mesh with low discretization error $\|u - u_h\|$.

3.2.2 Temporal Discretization

For unsteady part of equation (3.1), we find $u \in L_2(t_0, t_0+T; u_g + V(t))$

$$\frac{d}{dt}\int_\Omega uv\,dx + \int_\Omega \varepsilon \nabla u \cdot \nabla v + f(u)v\,dx = 0, \qquad \forall v \in V(t), \text{ and } t \in \Sigma, \tag{3.6}$$

where $V(t) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni u_g(t)|_{\Gamma_D} = g$. By using equation (3.4), equation (3.6) can be written in a more compact way with residual forms as follows:

$$\frac{d}{dt}m(u,v) + r(u,v) = 0 \quad \forall v \in V(t), \quad t \in \Sigma, \tag{3.7}$$

where $m(u,v) = \int_\Omega uv\,dx$ is the new temporal residual form.
We choose conforming finite element space $V_h^{k,d}(\mathscr{T}_h, t)$ and the function $u_{h,g}(t)$ depending on time so that $u_h(t) \in U_h(t) = u_{h,g}(t) + V_h(t)$. So, the spatial and temporal residual can be approximated as $r(u,v) \approx r_h(u_h, v_h)$ and $m(u,v) \approx m_h(u_h, v_h)$ respectively. Now for the integration of ODEs system, we subdivide the time interval into subintervals as follows

$$\overline{\Sigma} = \{t^0\} \cup (t^0, t^1] \cup \ldots \cup (t^{N-1}, t^N]$$

with $t^0 = t_0$, $t^N = t_0 + T$, $t^{k-1} < t^k$ for $1 \le k \le N$ and set the time-step as $\Delta t^k = t^{k+1} - t^k$. One of the simplest ODE integrators is the one-step-θ scheme that reads; find $u_h^{k+1} \in U_h(t^{k+1})$ such that

$$\frac{1}{\Delta t^k}(m_h(u_h^{k+1}, v; t^{k+1}) - m_h(u_h^k, v; t^k)) + \theta r_h(u_h^{k+1}, v; t^{k+1}) + (1-\theta) r_h(u_h^k, v; t^k) = 0$$
$$\forall v \in V_h(t^{k+1}).$$

Reordering the terms for $(k+1)$ and k shows that the method give solution to a nonlinear system [61], i.e., find $u_h^{k+1} \in U_h(t^{k+1})$ such that

$$m_h(u,v;t^{k+1}) + \Delta t^k \theta r_h(u,v;t^{k+1}) - m_h(u_h^k,v;t^k) + \Delta t^k (1-\theta) r_h(u_h^k,v;t^k) = 0 \quad \forall v \in V_h(t^{k+1}),$$

with $\theta = 1 - \sqrt{2}/2$. At each time level the algebraic system is solved by using BiConjugate Gradient(BiCG) method [56].

3.3 Results and Discussion

This section consists the two subsections. In first subsection, we simulate the problem for steady state and compute the L_2 error norm using uniform and adaptive grids strategies. In second subsection, we consider unsteady problem using different time discretization schemes. We compute the computational cost and temporal error for comparison with other schemes. All numerical simulations are preformed on Intel Core i3, 3.20×4 GHz CPU and 10 GB of RAM with Ubuntu 16.04 operating system.

3.3.1 Steady-state Solution

In this subsection, we consider (3.2) over domain $\Omega = (-5,5)^2$ and computed the error estimates on both uniform and adaptive grids using the exact solution [69]:

$$u(x,y) = \frac{\sinh\left(\sqrt{\frac{2}{\varepsilon}}x\right)}{\sqrt{2+\sinh^2\left(\sqrt{\frac{2}{\varepsilon}}x\right)}}, \tag{3.8}$$

and the exact solution (3.8) is taken as Dirichlet boundary data. We have used the value of $\varepsilon = 1.0$ and different degrees of freedom (DOF).

3.3.1.1 Uniform Grids

Computed results on uniform grids using the proposed method and its error distribution is illustrated in figure 3.1 [top to bottom] respectively. In the bottom row of figure 3.1, it can be seen that when we increase the number of grid points or degrees of freedom the error reduces. In bottom row of figure 3.1, L_2 error norm for; 80 degrees of freedom is 1.689×10^{-1} and 160 degrees of freedom is 1.22×10^{-2}. This concludes that minimization of error is seen by increasing degrees of freedom.

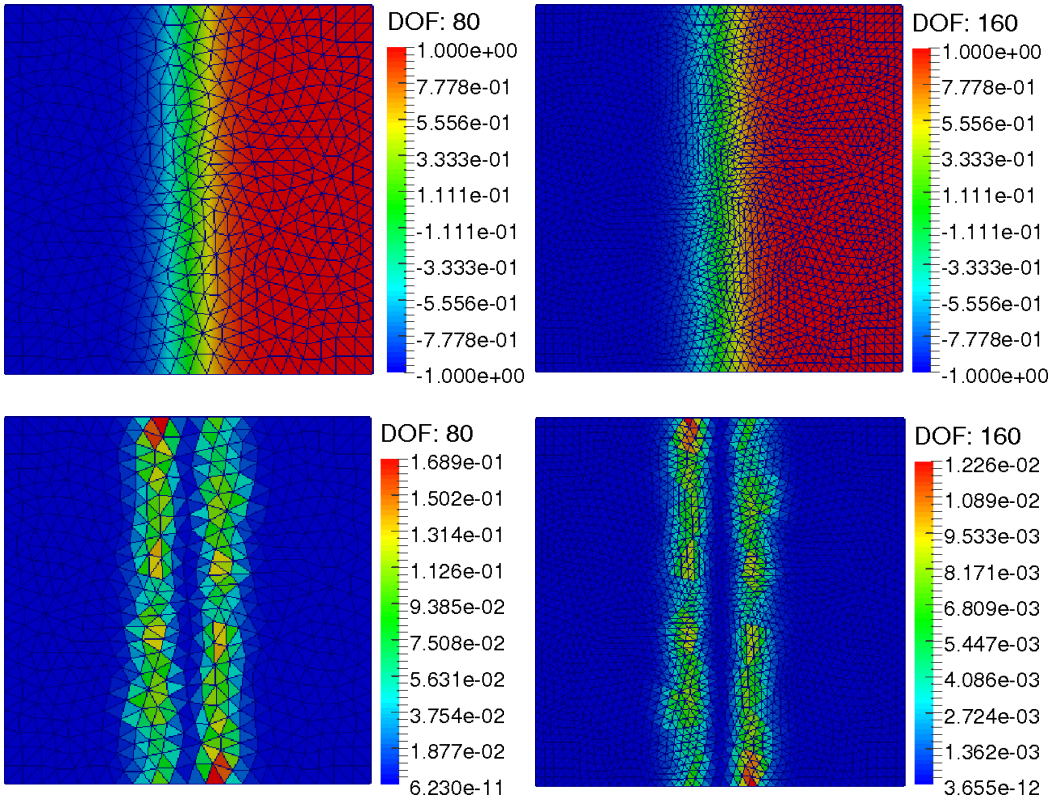


Figure 3.1: Uniform grids: FEM solution [top row], error distribution [bottom row]

3.3.1.2 Adaptive Grids

In this part, adaptive mesh refinement is performed to efficiently reduce the numerical error with minimal computational cost and clustering of grids near the interface. Figure 3.2 illustrate the computed solution [top row] at 66 and 122 DOF and its error distribution [bottom row]. When we compare the L_2 error norm of adaptive grids with uniform and the bottom rows of figures 3.1 and 3.2 show that the computed L_2 error norm is low with adaptive grids even for low degrees of freedom. In adaptive grids, we use the more grids points near the interface where the phenomenon has jump discontinuities and also for less degrees of freedom, the better results can be achieved as compared to uniform grids which can reduce the computational cost and memory.

In Table 3.1, comparison of L_2-error estimate using adaptive and uniform grids at different degrees of freedom are given. We can see the advantage of using adaptive grids near the regions of high gradient.

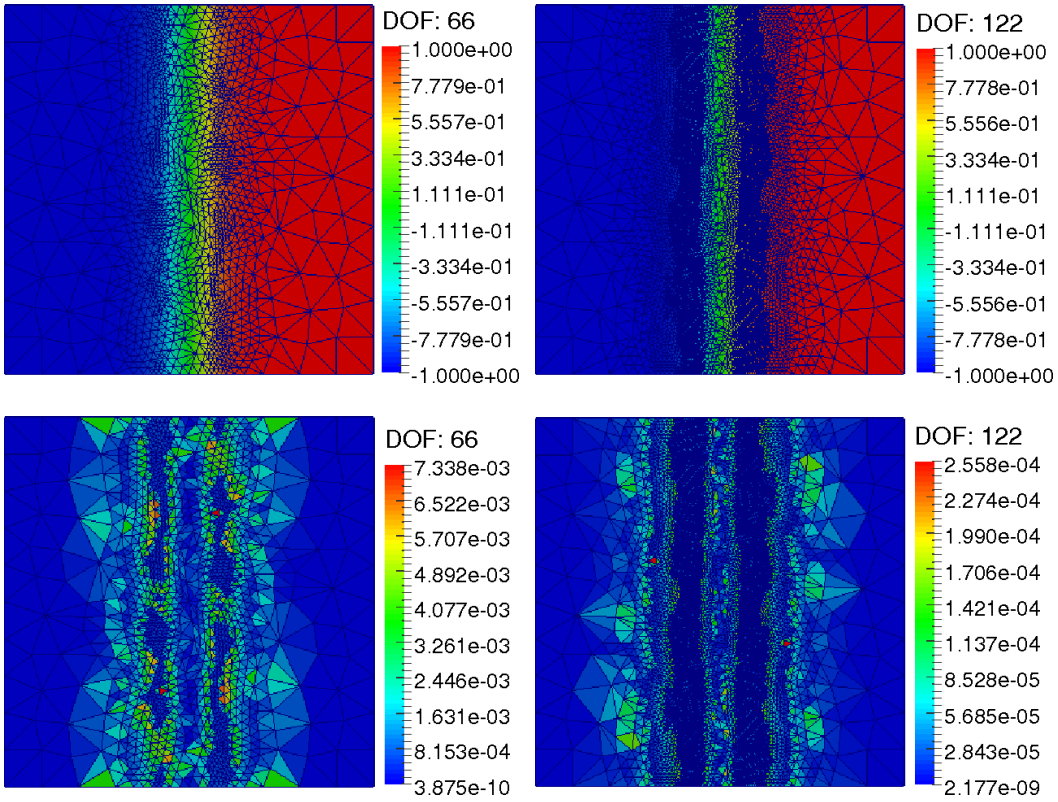


Figure 3.2: Adaptive grids: FEM solution [top row], error distribution [bottom row]

DOF		Elements		L_2-error	
Adaptive grids	Uniform grids	Adaptive grids	Uniform grids	Adaptive grids	Uniform grids
66	80	1287	537	7.338e-03	1.689e-01
122	160	7373	2065	2.558e-04	1.226e-02
298	320	43807	8097	4.435e-06	7.821e-04
416	640	113369	32065	7.342e-07	4.909e-05
1114	1280	677267	127617	2.040e-08	3.071e-06
1600	2560	1775147	509185	3.229e-09	1.920e-07

Table 3.1: Comparison of adaptive and uniform grids at different degrees of freedom

3.3.2 Unsteady Solution

This subsection provides the unsteady solution of equation (3.1). This subsection is divides into three parts. Firstly, we investigate the applicability of our algorithm to solve time-dependent problems. By using equation (3.8) as initial condition, we have solved equation (3.1) for $t = 0$ to $t = 1.0$ with time-step $\Delta t = 0.1$ as shown in figure 5.3. A stable solution is observed for $\varepsilon = 1.0$ and $\varepsilon = 0.02$.

Secondly, we provides the comparative study of our algorithm with some other schemes. We have used the explicit schemes (Shu's and Heun's), implicit Euler scheme and

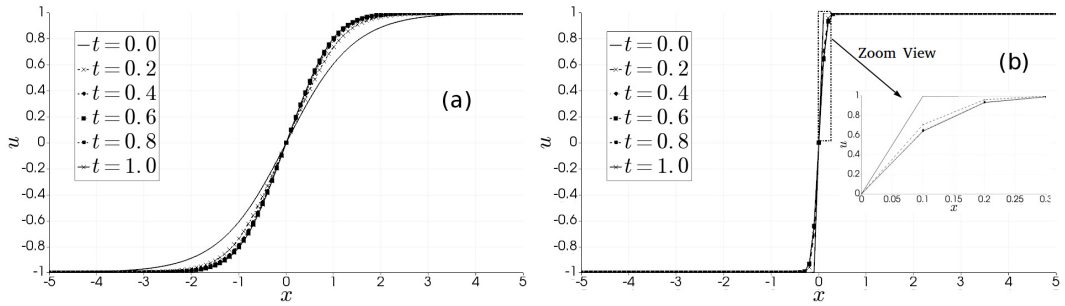


Figure 3.3: Unsteady solution at different time levels with: (a) $\varepsilon = 1.0$ (b) $\varepsilon = 0.02$.

Alexsender's scheme (DIRK2) for comparison with our DIFST scheme. We consider the one-dimensional problem over domain $\Omega = (-10, 10)$ with mesh size $h = 0.009765625$ and $\varepsilon = 0.02$ by using the equations (3.8) and (1.2) as initial and boundary conditions respectively. We preformed the numerical experiments with different Δt from $t = 0.0$ to $t = 1.0$. Table 3.3.2 shows the comparison of computational cost in term of CPU time. From Table 3.3.2, it is clear that the proposed DIFST scheme is robust. From Table it is also clear that the explicit schemes are unstable for last time-step.

Schemes	$\Delta t = 10^{-4}$	$\Delta t = 10^{-3}$	$\Delta t = 10^{-2}$	$\Delta t = 10^{-1}$
DIFST	59.636	18.231	2.4540	0.27976
DIRK2	61.312	18.702	2.6001	0.29508
Euler's	72.091	20.316	3.7192	0.59310
Heun's	60.234	Fails	Fails	Fails
Shu's	63.631	Fails	Fails	Fails

Table 3.2: Comparison of computational cost for proposed scheme with some other schemes.

Finally, we have computed the local temporal error for above said schemes. Figure 3.4(a) shows the local temporal error for Shu's and Heun's explicits schemes and figure 3.4(b) for implicit (Euler, DIRK2 and DIFST). Here we can see that the explicit scheme have low error for small time-step size.

Figure 3.5(a-c) shows the local temporal error comparison among three implicit schemes. From figure 3.5(a-c), it is clear that the DIFST scheme have less error as compared to DIRK2 and implicit Euler.

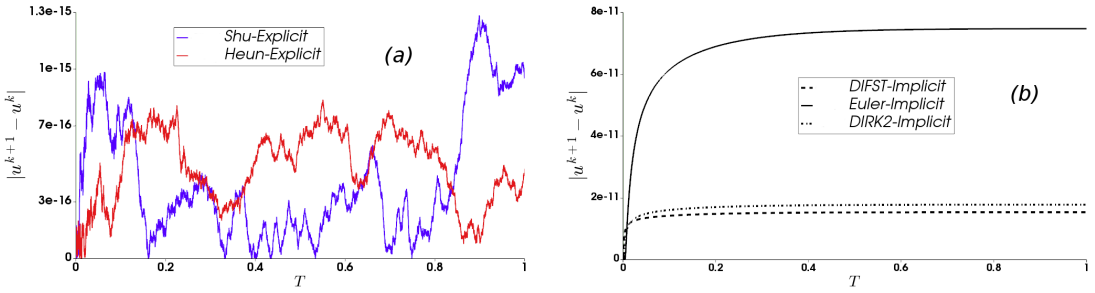


Figure 3.4: Local temporal error comparison among (a) explicit schemes and (b) implicit schemes

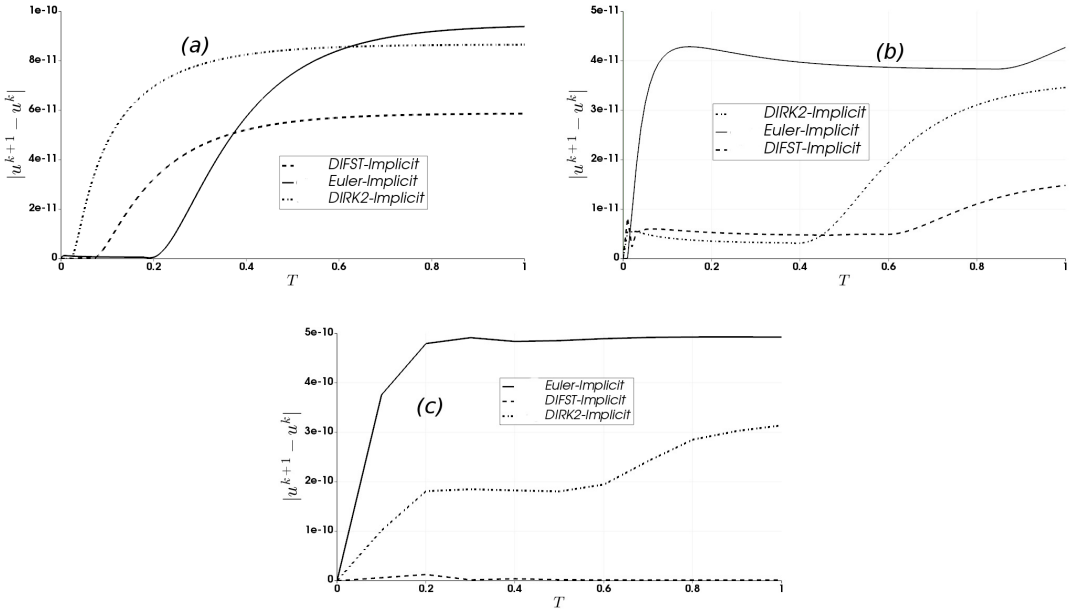


Figure 3.5: Comparison of DIFST scheme with other implicit and diagonally implicit schemes for large Δt.

3.3.3 Test Problems

In this subsection, we simulate some benchmark test problem to check capability the numerical scheme for Allen-Cahn dynamics for moving interface problem. We consider the motion of a circular interface [73] over the domain $[-5,5]\times[-5,5]$ with $\varepsilon = 0.02$ by imposing the homogeneous Drichlet and Neumann boundary conditions. Firstly, we take the initial condition [74]:

$$u_0(x,y) = \begin{cases} 1, & x^2+y^2 \leq 2 \\ -1, & \text{otherwise.} \end{cases}$$

From figure 3.6, we can see that due to the non-conservative nature of the Allen-Cahn equation, the circular region start shrinking with time and finally disappear at $t = 25$.

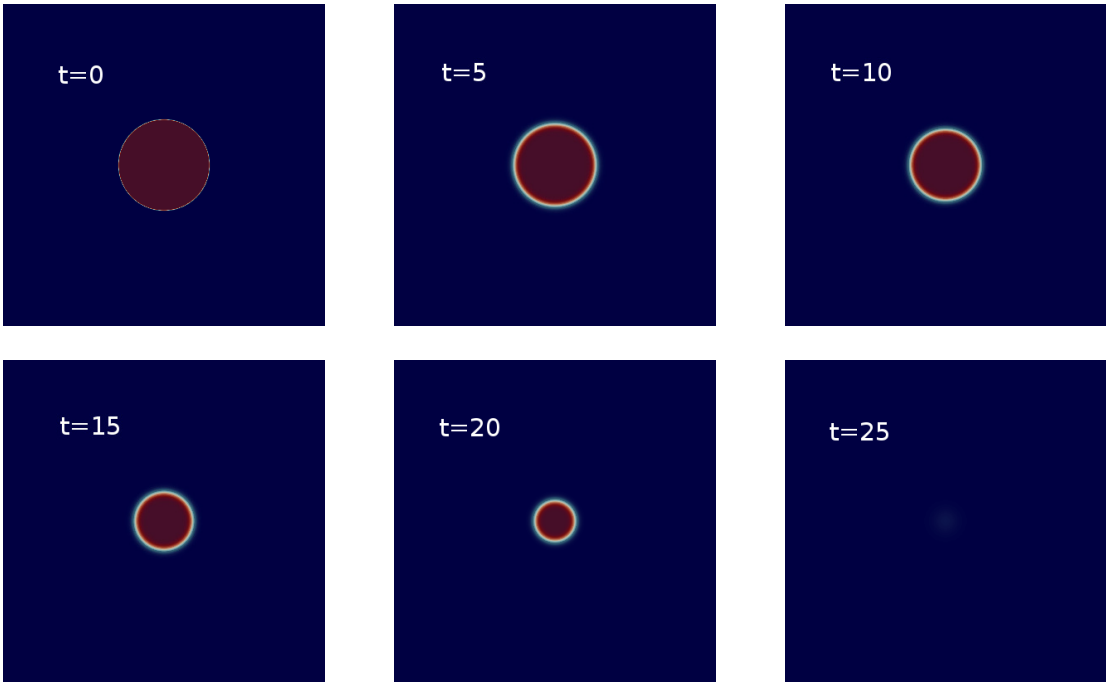


Figure 3.6: Snapshots for shrinking of circular bubble at different times

Secondly, we consider square shape with following initial condition:

$$u_0(x,y) = \begin{cases} 1, & |x| \leq 2 \ \& \ |y| \leq 2 \\ -1, & \text{otherwise,} \end{cases}.$$

and simulated for $t = 40$. We can observe that with of time, square shape deforms into a circular shape as wel as shrinks as shown in figure 3.7 and will vanish which is observed from figure 3.6.

Next, we have simulated coalescence two bubbles with following initial condition [74]:

$$u_0(x,y) = \begin{cases} 1, & (x+2)^2 + y^2 \leq 2 \ \& \ (x-2)^2 + y^2 \leq 2 \\ -1, & \text{otherwise.} \end{cases}$$

Due to involvement of time, in the beginning two bubbles coalesce into a elliptical bubble, which deforms into a steady state as one circular bubble as shown in figure 3.7. It also clear from figure 3.7, the over all mass of two bubbles is decreasing and deforming into circular shape.

To keep the mass conserve, either we need to solve the volume preserving Allen-Cahn equation or Cahn-Hilliard equation [75]. One of the reasons for choosing Allen-Cahn equation is due to its ease in numerical treatment than that of the Cahn-Hilliard type which involves fourth-order differential operators. Due to non-conservative property of

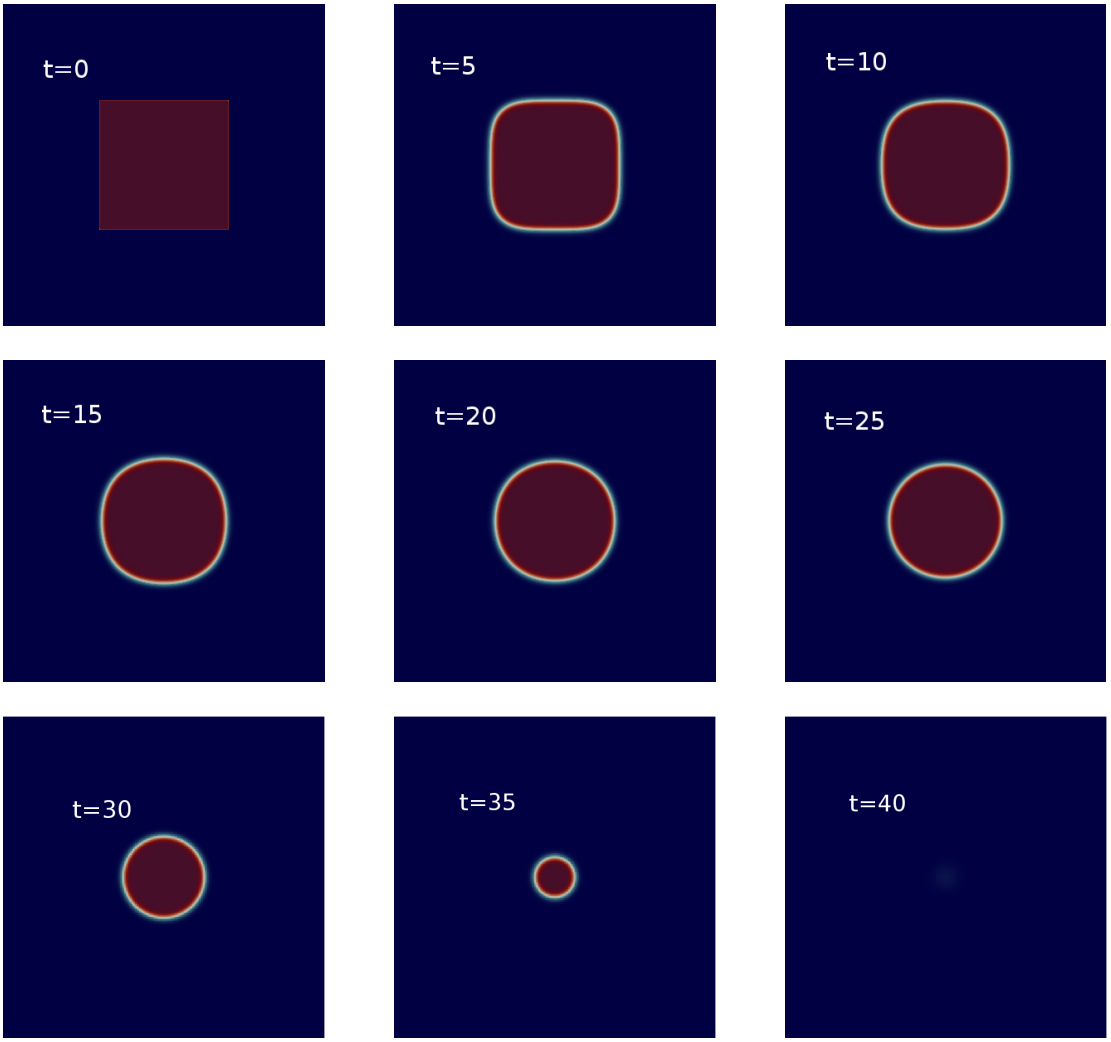


Figure 3.7: Snapshots for deformation square bubble into circular bubble at different times

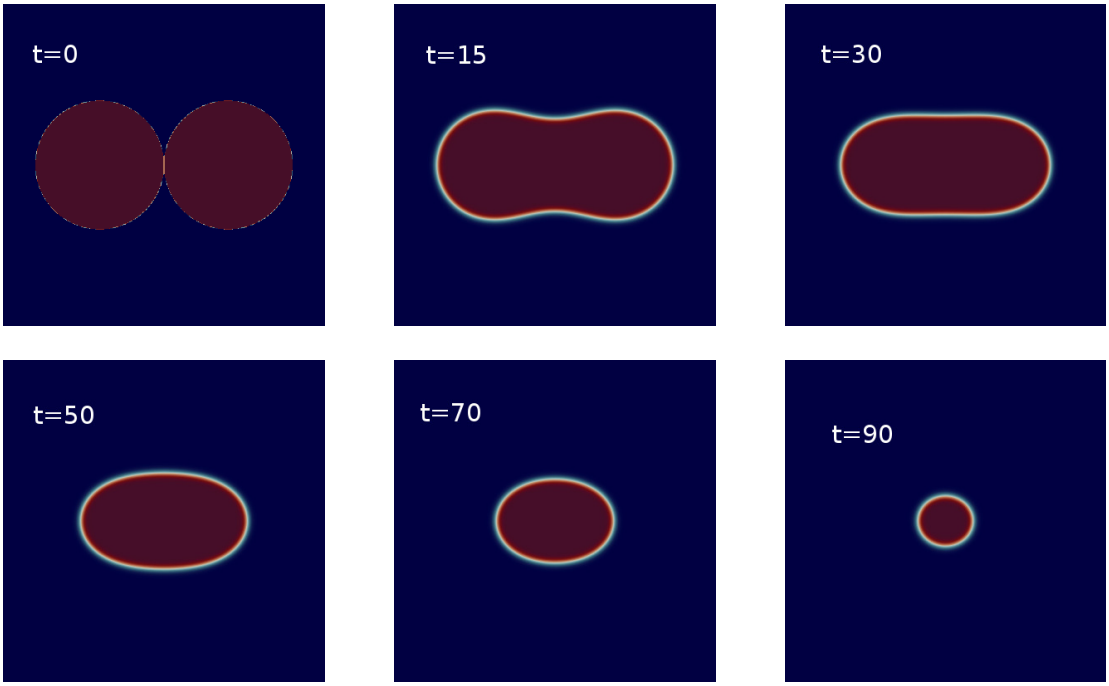


Figure 3.8: Snapshots for coalescence of two circular bubbles at different times

Allen-Cahn phase-field model, the volume will decrease [76]. Figure 3.9 shows the rate of volume loss in term of volume fraction versus time.

3.4 Summary

We have proposed an efficient and robust algorithm for the numerical simulation of Allen-Cahn equation. Conforming P_1 FEM is used for space discretization while diagonally implicit fractional step θ-scheme is used for time discretization. For steady state solution, adaptive and uniform grids are used and effectiveness of adaptive grids over uniform grids has been studied by computing the L_2 error. For unsteady solution, the efficiency and robustness of diagonally implicit fractional step-θ scheme is illustrated by comparing with some other schemes in term of computational cost and local temporal error. Furthermore, the simulation of an unsteady test problem is provided to show the capability of method for simulating time-dependent problems for large time-step Δt.

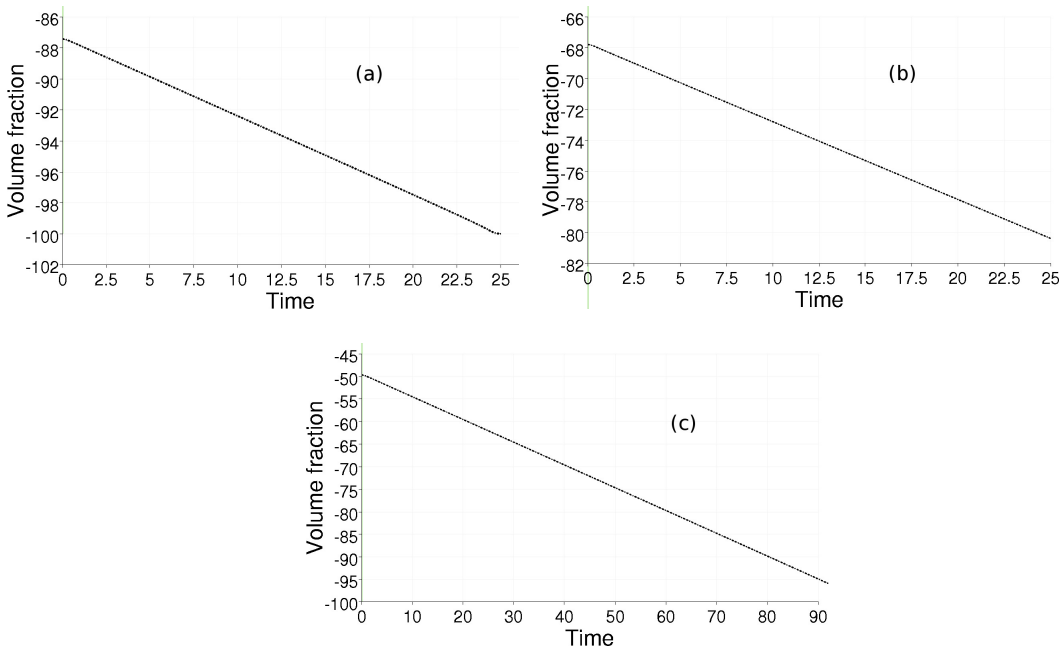


Figure 3.9: Volume fraction versus time: (a) circular bubble, (b) square and (c) coalescences of two bubbles

Chapter 4

Numerical Solution of Cahn-Hillard Equation with Variable Mobility

In this chapter, we try to develop an adaptive time-stepping numerical scheme for simulation of Cahn-Hillard equation with variable mobility. We simulate the phenomenon of chemical separation known as spinodal decomposition. Spinodal decomposition is a mechanism for the rapid unmixing of a mixture of liquids or solids from one thermodynamic phase to form two coexisting phases. We consider the 6th-order double well potential along with 4th-order variable mobility function. The proposed numerical method is validated for solving some 1-D test problems. Finally, the algorithm is applied to simulate 2D spinodal decomposition phenomena.

4.1 Introduction

The Cahn-Hilliard equation is used to study many physical processes in applied science and engineering from nano/micro-scale to the chunking of galaxies in universe [77, 78]. Also the phase transition phenomenon in homogeneous mixture known as spinodal decomposition can be studied using this equation. We consider the Cahn-Hillard equation (1.12) introduced in chapter 1 as:

$$\frac{\partial u}{\partial t} = \nabla.[M(u)\nabla c], \tag{4.1}$$

$$c(\mathbf{x},t) = F^{'}(u) - \varepsilon^2\nabla^2 u. \tag{4.2}$$

where $F(u) = u^6 - u^4 - u^2 + 1$ is double-well potential [79] and $M(u) = \frac{1}{4}(u^2-1)^2$ is the variable mobility which may be a constant or concentration $u(\mathbf{x},t)$ dependent [80]. The following initial condition

$$u(x,y) = u_0(x,y) \qquad \text{at } t = 0. \tag{4.3}$$

and no-flux boundary conditions

$$\nabla u \cdot \nu = \nabla c \cdot \nu = 0 \qquad \text{on } \partial\Omega. \tag{4.4}$$

are used to solve equation $(4.1-4.2)$.

The Cahn-Hilliard equation has been studied numerically with constant mobility [81, 82, 83] and with concentration dependent variable mobility [84, 85, 86]. Many time-discretization schemes (e.g backward -Euler, semi-implicit and higher order implicit Runge-Kutta etc.) have been used for numerical simulation of the equation [34, 35]. The aim of this work is to develop an accurate method for solving Cahn-Hilliard equation with variable mobility. For this purpose, we adapted the P_1 conforming FEM

[70, 71] and diagonally implicit fractional step θ-scheme (DIFST) [62] for spatial and temporal discretization respectively.

4.2 Numerical Discretization

Finding the analytical solution of equations $(4.1-4.2)$ is not easy task due to its nonlinear nature, therefore we have developed numerical method based on the P_1 conforming FEM [70, 71] for spatial discretization and DIFST scheme for temporal discretization [62].

4.2.1 Space Discretization

In finite element method, the following weak form of the steady-state Cahn-Hilliard equation $(4.1-4.2)$ can be obtained using the test function $v \in H_0^1(\Omega)$ and imposing no-flux boundary conditions given in equation (4.4). For this find $u \in U$(trail function space) such that the spatial residual $r(v,u) = 0 \quad \forall v \in V$(test function space),

$$\begin{aligned} r(v,u) &= a(v,Mc), && (4.5)\\ (v,c) &= (v,f^{'}(u)) + a(v,\varepsilon^2 u), && (4.6) \end{aligned}$$

where $(.,.)$ and $a(.,.)$ are L_2 and energy inner product on Ω respectively. Next, define $u^h \in U^h$ (h being mesh size) to be the finite dimensional approximation of concentration field u, $c^h \in W^h$ to be the finite dimensional approximation of chemical potential field c and $v^h \in V^h$ to be the finite dimensional test function are given as follows:.

$$\begin{aligned} U^h &= \left\{ u^h | u^h \in H_0^1,\ u^h \in P^1(\Omega) \right\},\\ W^h &= \left\{ c^h | c^h \in H_0^1,\ c^h \in P^1(\Omega) \right\},\\ V^h &= \left\{ v^h | v^h \in H_0^1,\ v^h \in P^1(\Omega) \right\}. \end{aligned}$$

An approximate solution to the split Cahn-Hilliard equation is to find $u^h \in U^h$ and $c^h \in W^h$ such that:

$$\begin{aligned} r_h(v^h,u^h) &= a_h(v^h,Mc^h), && (4.7)\\ (v^h,c^h) &= (v^h,f^{'}(u^h)) + a_h(v^h,\varepsilon^2 u^h) \qquad \forall v^h \in V^h, && (4.8) \end{aligned}$$

where: $v^h = \sum^n vP^1$, $u^h = \sum^n uP^1$ and $c^h = \sum^n cP^1$ with P^1 as basis function and n as the dimension of discrete space.

4.2.2 Time Discretization

There are two main components of efficient time discretization schemes: choice of temporal scheme and adaptivity. Due to stiff nature (small interfacial length) of the Cahn-Hilliard equation, the explicit methods have stability problem. Therefore, implicit methods becomes natural alternative due to their unconditional stability. Probably the most utilized implicit time schemes in this context are Euler backward and Crank-Nicholson methods [36, 37, 40]. The explicit-implicit time-stepping algorithms are an intermediate approaches which can be applied only to problems with constant mobility. Recently methods based on implicit Runge-Kutta schemes are proposed to solve Cahn-Hilliard equation with variable mobility [35, 28]. We used an adaptive implicit time stepping scheme to simulate multi-scale phenomena with variable mobility in optimal computational cost. In this work, we develop an efficient adaptive time-stepping scheme for solving Cahn-Hilliard equation with variable mobility using diagonally implicit fractional step θ-scheme (DIFST) [62]. We consider the equation $(4.1-4.2)$ for unsteady problem and wrote the new temporal residual along with spatial residual as in equation (4.7):

$$\frac{d}{dt} m_h(v^h, u^h) + r_h(v^h, u^h) = 0. \tag{4.9}$$

For equation (4.9), subdivide the time interval into not necessarily equidistant subintervals:

$$\overline{\Sigma} = \{t^0\} \cup (t^0, t^1] \cup \ldots \cup (t^{N-1}, t^N]$$

with $t^0 = t_0$, $t^N = t_0 + T$, $t^{k-1} < t^k$ for $1 \leq k \leq N$ and set the time step to $\Delta t^k = t^{k+1} - t^k$. For θ-scheme we have to find $u_h^{k+1} \in U_h(t^{k+1})$, s.t.:

$$\frac{1}{\Delta t_k}(m_h(u_h^{k+1}, v; t^{k+1}) - m_h(u_h^k, v; t^k)) + \theta r_h(u_h^{k+1}, v; t^{k+1}) + (1-\theta) r_h(u_h^k, v; t^k) = 0, \quad \forall v \in V_h(t^{k+1}). \tag{4.10}$$

We apply diagonally implicit fractional step θ-scheme which is suitable for nonlinear

and stiff (free boundary) problems [62]. At each time level, a sparse linear system is required to solve using SuperLU method [63].

To reduce the computational cost, we provide an adaptive time-stepping algorithm. For this let u_h^k and c_h^k be approximate solutions of equation $(4.7-4.8)$ and we set $t^{k+1} = t^k + \sigma\Delta t^k$ and Δt^k is the nonuniform time step size with $\sigma = 1.1$ being an adaptive factor. In previous studies the following time steps were used in the numerical simulations of the Cahn-Hilliard equation: $\Delta t = 10h^2$ [45], $\Delta t = h^2 \sim 4h^2$ [87], $\Delta t = 0.5h^2$ [88], $\Delta t = 25h^{2/3}$ [83], and $\Delta t = h^2$ [86]. Recently Kim et.al [33] have used an adaptive time-stepping method for numerical simulations of the Cahn-Hilliard equation with constant mobility. However, we use $\Delta t_{min} = h^2$ and $\Delta t_{max} = 100\Delta t_{min} \sim 1000\Delta t_{min}$. The adaptive time-stepping algorithm is given as follows:

Algorithm 1 Adaptive time-stepping process

Given: $\mathbf{U^k}$ and Tolerance: Tol, $\Delta t_{\min} = 10^{-5}$, $\Delta t_{\max} = 100\Delta t_{\min} \sim 1000\Delta t_{\min}$, End Time $= t$

while Time $<$ t **do**,

  Compute $\mathbf{U^{k+1}}$ using $\Delta t^k = \Delta t_{\min}$

  Calculate $e_{k+1} = |\mathbf{U^{k+1}} - \mathbf{U^k}|$

  if $e_{k+1} >$ Tol **then**

    $\Delta t^k = \Delta t^k/2$

    Recompute $\mathbf{U^{k+1}}$,

  else

    Update the time step.

    If $\Delta t^k < \Delta t_{\max}$

    Then $\Delta t^k = \min(\sigma\Delta t^k, \Delta t_{\max})$; where σ is adaptive factor.

  end if

end while

4.3 Results and Discussion

We perform numerical experiments to show the ability of Algorithm 1. We also aim to find the largest time-step (Δt) for which the algorithm is stable relative to value of interfacial width ε. We simulated three different phenomena and computed the CPU

time.

4.3.1 Interface Capturing

By considering $u_0(x,t) = tanh(\frac{x}{\sqrt{2}\varepsilon})$ as an initial condition, we have simulated the problem for $\varepsilon = 0.1$ to $\varepsilon = 0.01$ at $t = 1.0$ as shown in figure 4.1 and have noted the CPU time in seconds for $t = 1.0$ given in Table 4.1.

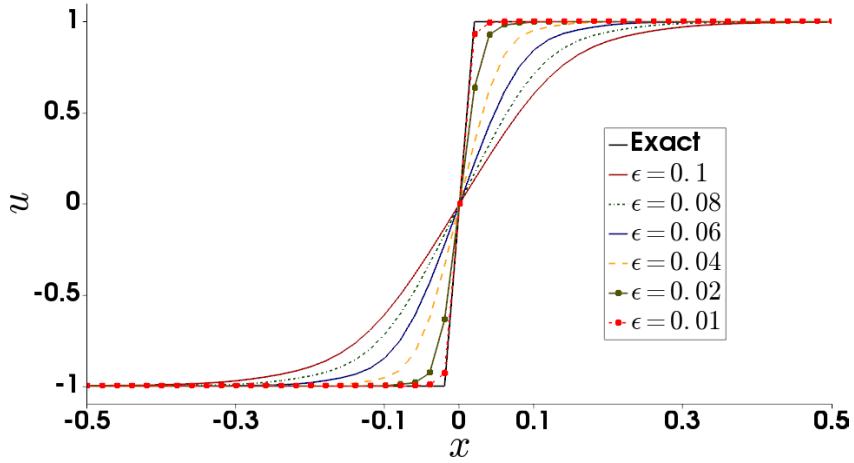


Figure 4.1: Solution profiles at different values of ε at $t = 1.0$

It can be observed that the algorithm is capable to solve the interfacial problem for very small interface width. In Table 4.1, we have shown the effects of interfacial width on CPU time and observed that decreasing ε, results in increase of CPU time.

Δt_{min}	Δt_{max}	ε	CPU Time
10^{-5}	10^{-2}	0.1	4.5608
10^{-5}	10^{-2}	0.08	4.6506
10^{-5}	10^{-2}	0.06	4.8342
10^{-5}	10^{-2}	0.04	5.0159
10^{-5}	10^{-2}	0.02	5.0228
10^{-5}	10^{-2}	0.01	5.0686

Table 4.1: Effects of interfacial width on CPU time

4.3.2 Computational Cost

In this subsection, we have performed the two test cases with sinusoidal perturbation for $t = 0.6$ (case-1) and $t = 2$ (case-2) with $h^2 = 10^{-5}$ over the domain $\Omega = (0,1)$ having fixed value $Tol = 2 \times 10^{-6}$ and interfacial width $\varepsilon = 0.01$.

4.3.2.1 Case-1

In this case, we consider an initial condition as [33]:

$$u_0(x,t) = 0.1cos(20\pi x) + 0.1cos(30\pi x^2) + 0.1cos(40\pi x^3).$$

Table 4.2 shows that by increasing Δt_{max}, CPU time is decreased. In Table 4.2, the top row shows the computational cost (CPU time) for uniform time step size, we can see here that the computational cost is extensively high as compared to adaptive time step. In Table 4.2 last column indicates the saving of time in percentage % with adaptive time stepping.

Δt_{min}	Δt_{max}	Steps	CPU Time	Time Saving in %
10^{-5}	10^{-5}	60000	1142.4	
10^{-5}	1×10^{-3}	639	9.8986	99.13352591
10^{-5}	2×10^{-3}	346	5.4006	99.527258403
10^{-5}	3×10^{-3}	250	3.8665	99.661545868
10^{-5}	4×10^{-3}	203	3.1506	99.724212185
10^{-5}	5×10^{-3}	176	2.7804	99.756617647
10^{-5}	6×10^{-3}	158	2.5444	99.77727591
10^{-5}	7×10^{-3}	145	2.4107	99.788979342
10^{-5}	8×10^{-3}	136	2.2275	99.805015756
10^{-5}	9×10^{-3}	129	2.1467	99.812088585
10^{-5}	1×10^{-2}	123	2.0513	99.820439426

Table 4.2: Comparison of CPU time among constant and adaptive time steps for case-1

4.3.2.2 Case-2

Next, we consider the following initial condition [33]:

$$u_0(x,t) = 0.1sin(2\pi x) + 0.01cos(4\pi x) + 0.05sin(4\pi x) + 0.02cos(10\pi x).$$

We have computed the computational cost with uniform and adaptive time steps and compared and given in Table 4.3. The first row of Table 4.3 lists CPU time in seconds for uniform time step size case and the last column indicates the saving of time in percentage % with adaptive time step. It is clear that from CPU time for adaptive time step is less than uniform time step.

Δt_{min}	Δt_{max}	Steps	CPU Time	Time Saving in %
10^{-5}	1×10^{-5}	100000	1977.2	
10^{-5}	1×10^{-3}	2039	31.6	98.401780295
10^{-5}	2×10^{-3}	1046	16.24	99.178636456
10^{-5}	3×10^{-3}	717	11.099	99.438650617
10^{-5}	4×10^{-3}	553	8.4777	99.571226988
10^{-5}	5×10^{-3}	456	7.0176	99.645073842
10^{-5}	6×10^{-3}	391	5.9812	99.697491402
10^{-5}	7×10^{-3}	345	5.2529	99.73432632
10^{-5}	8×10^{-3}	311	4.7644	99.759032976
10^{-5}	9×10^{-3}	284	4.4489	99.774989885
10^{-5}	10×10^{-3}	263	4.2284	99.786142019

Table 4.3: Comparison of CPU time among constant and adaptive time steps for case-2

4.3.3 2-D Simulation of Spinodal Decomposition

In the spinodal decomposition simulation, initial separation is fast process but later the coarsening is slow. To address this multi-scale behavior, we utilize adaptive time-stepping based on rigorous error control. The computational domain of $\Omega = [0,1]\times[0,1]$, $h = \frac{1}{316}$, $tol = 2\times10^{-6}$, $\Delta t_{min} = h^2$, $\Delta t_{max} = 1000h^2$ and interfacial width $\varepsilon = 0.01$ are used. The initial condition is

$$u_0(x,y,0) = 0.02rand(x,y),$$

where $rand(x,y)$ is a random number generator between 1 and -1.

Figure 4.2(left to right and top to bottom) shows the initial behavior of spinodal decomposition phenomena which represents the fast dynamics. In each stage the time-step size play crucial role due to sudden changes in morphology. Starting from $\Delta t_{min} = h^2$

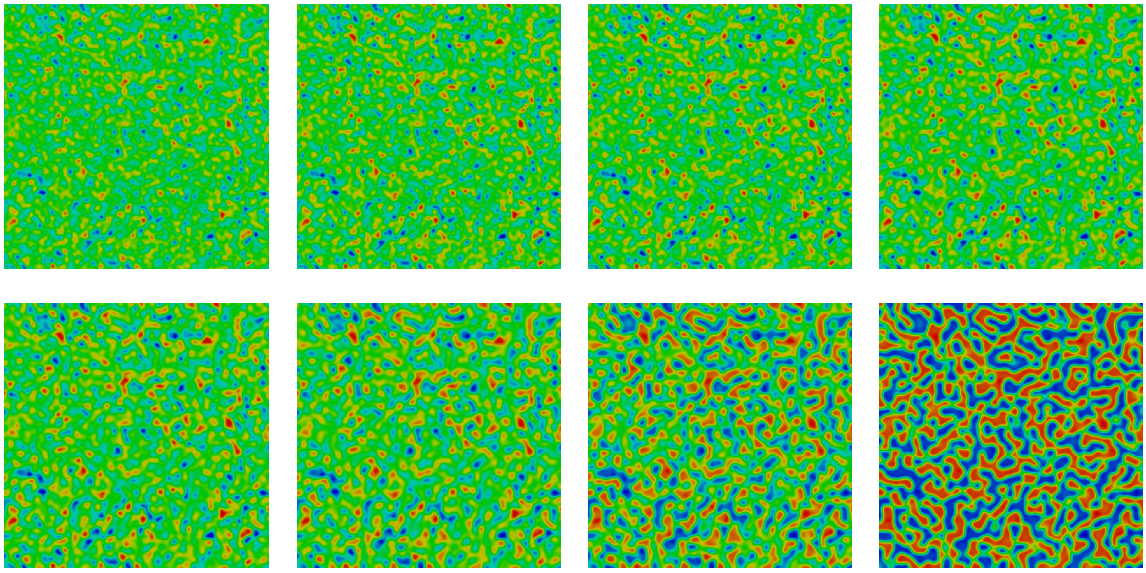


Figure 4.2: Fast dynamical behavior of chemical separation from $t = 0.0$ to $t = 0.10502$ (left to right and top to bottom)

t	Δt_{max}	Steps	CPU Time
1×10^{-5}	1×10^{-5}	1	40.372
2.4523×10^{-4}	3.1384×10^{-5}	13	384.17
7.9543×10^{-4}	8.1403×10^{-5}	23	664.18
2.2225×10^{-3}	2.1114×10^{-4}	33	939.93
5.9240×10^{-3}	5.4764×10^{-4}	43	1174.5
1.5525×10^{-2}	1.4204×10^{-3}	53	1440.5
4.0427×10^{-2}	3.6842×10^{-3}	63	1676.0
1.0502×10^{-1}	9.5559×10^{-3}	73	1987.1

Table 4.4: Computational cost for fast dynamics

with fixed $\Delta t_{max} = 1000 \times h^2$, 73 time-steps iteration with adaptive factor σ are performed from $t = 0.0$ to $t = 0.10502$ as shown in figure 4.2 (left to right and top to bottom). Table 4.4 shows the CPU time in seconds and time steps which needed to simulate phenomenon for $t = 0.1$

Figure 4.3. (left to right and top to bottom) shows temporal evolution of morphological patterns between initial phase separation and coarsening at $t = 0.2$ to $t = 0.9$. Table 4.5 shows the CPU time in seconds and time steps which needed to simulate phenomenon for $t = 0.2$ to $t = 0.9$.

Figure 4.4 (left to right and top to bottom) shows morphological patterns during coars-

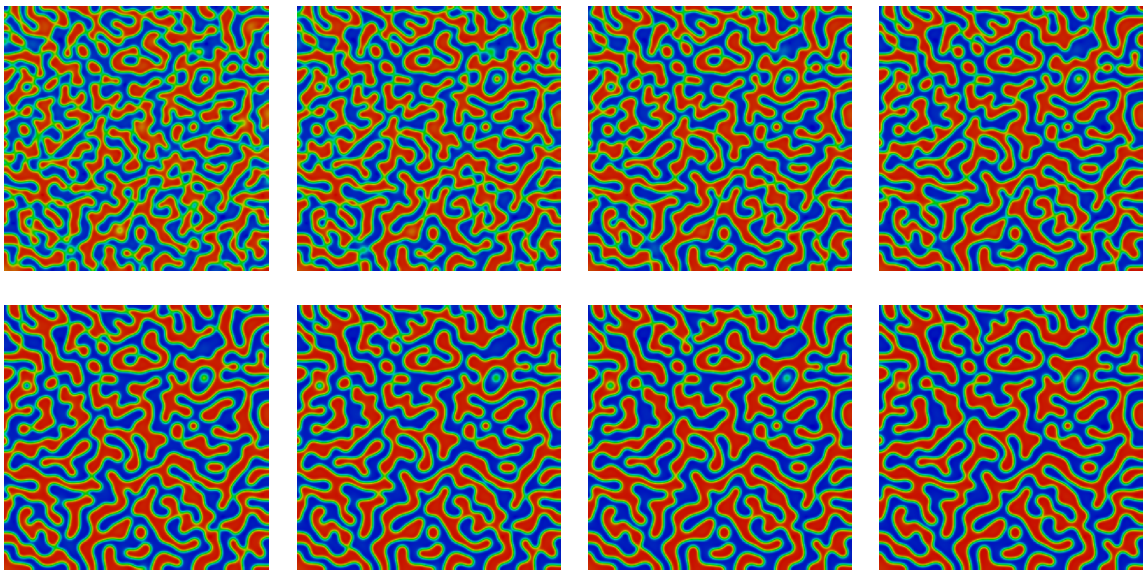


Figure 4.3: Behavior of chemical separation from $t = 0.10502$ to $t = 0.9$ (left to right and top to bottom)

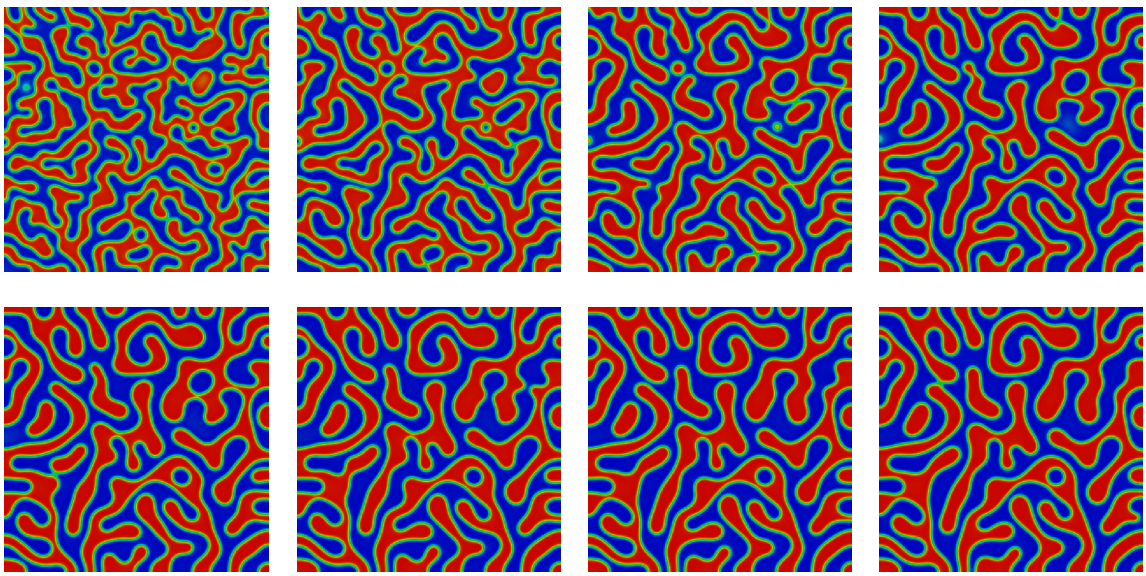


Figure 4.4: Coarsening behavior of chemical separation from $t = 1.0$ to $t = 8.0$ (left to right and top to bottom)

t	Δt_{max}	Steps	CPU Time
0.2	10^{-2}	83	4209
0.3	10^{-2}	93	6352
0.4	10^{-2}	103	9090
0.5	10^{-2}	113	11938
0.6	10^{-2}	123	14728
0.7	10^{-2}	133	17530
0.8	10^{-2}	143	20839
0.9	10^{-2}	153	24344

Table 4.5: Computational cost from fast dynamics to coarsening process

t	Δt_{max}	Steps	CPU Time
1	10^{-2}	163	4209
2	10^{-2}	263	6352
3	10^{-2}	363	9090
4	10^{-2}	463	11938
5	10^{-2}	563	14728
6	10^{-2}	663	17530
7	10^{-2}	763	20839
8	10^{-2}	863	24344

Table 4.6: Computational cost for coarsening process

ening which allows to use the large time-step. Now our time-step size has been increased up to $1000 \times h^2 = 10^{-2}$ which reduces computational cost. We obtain the better result in figure 4.4 with only 873 time-step and previously, was obtained with 3077 [33] and 30000 time-steps [86]. Also table 4.6 shows the CPU time in seconds and time steps which needs to simulate phenomenon for $t = 1.02$ to $t = 8.0$.

4.4 Summary

We have proposed an adaptive time-stepping scheme for numerical solution of Cahn-Hilliard equation with variable mobility using the method developed in chapter 2. Some test cases are computed to illustrate the advantage of adaptive time-stepping approach.

Also a 2D problem of spinodal decomposition with random initial condition is simulated and computed results are provided graphically.

Chapter 5

Numerical Simulation of Dendritic Crystal Growth

This chapter deals with the numerical simulation of phase-field model for dendritic crystal growth phenomena. Effects of some parameters on the growth and shape of dendrites are presented and comparison with other numerical methods is provided.

5.1 Introduction

Due to new development in material science research, the study of dendritic crystal growth is an essential phenomena which occurs in binary alloys. In the such type of crystals growth process, the temperature of the system has significant effects on the micro-stucture of the crystals [89]. During the recasting of metals and alloys, these micro-structure of the crystals grow in solidification process with expense of liquid phase into solid phase [90]. The study of size and shape of these micro-structure with the passage of time is very nonessential because during this process the material properties are being changed. However, the topological structure of formation of dendritic crystals is not still fully understood phenomenon [91]. Therefore, modeling of complex phenomenon of dendritic crystal growth are very challenging in applied sciences. On one hand, the micro-structures in phase formation are developed which goes to the process of solidification in thermo-mechanical procedure. These procedures are based on the dynamics of the mechanical limitations of the kinetics of non-equilibrium and free boundary. Small seeds grow in solidification process and reduce the effect of the free surface energy of the system, is the result of mass and heat diffusion. On the other hand, solidification/freezing typically serves as a model with several phases of transition modes and result, the formation of asymmetric complex micro-structures. Patterning is well known because they look solid at room temperature and can happen during the process of solidification. However, dendrite materials are rich rather caustic and therefore vulnerable [23]. Initially, the idea to simulate two-phase flow problems with jump/sharpness at interface was introduced by Fix [92]. In material science research, the problems where the interface has varying behavior due to different material properties called free/moving boundary problems. However, the analytical or numerical solution of such problems having sharp edges/discontinuities at interface is difficult task. An alternative of sharp interface model like level set and phase-field methods [93]. The phase-field method is better choice where the sharp interface is replaced by diffusive region of small width called interfacial width. Recently, the phase-field models have

became better choice in computational material sciences due to inherited simplicity at the interface. The phase-field mathematical model having coupled system of partial differential equations (PDE's) was first used by Kobayashi [22] and Levine et al. [94] for simulation the dendrite crystal growth phenomena of solidification under-cooling. In the modeling of many physical systems where two or more different processes are combined, stiff PDE's are arises. Numerical methods for solving these PDE's need more work per step. Explicit method perform poorly for these equations where as the implicit methods is the better choice. In implicit methods, matrix operation is used to solve a system of simultaneous linear equations at each step which helps to predict the evolution of the physical process. Alexander [61] solved stiff problems using diagonally implicit Runge-Kutta (DIRK) methods of different order of accuracy. The phase-field model for free boundaries consists of a system of parabolic differential equations in which the variables represent a phase parameter and diffusion of temperature. Recently an extensive effort are made by Jimack's et. al [95] in developing numerical techniques for the simulation of dendrite growth phenomenon of solidification with focus on mesh size, its refinement and de-refinement. In view of the work done by Jimack, it still needed to develop an efficient time discretization schemes which allow the simulation of solidification process for large time step to reduce the computational cost.

5.2 Mathematical Model

The phase-field modeling is a mathematical representation for two-phase gradient flow by replacing the sharp interface between two different phases by the non-zero transition layer [96] with interfacial thickness ε and defining phase function $u(\mathbf{x},t)$ which varies from 0 to 1. For example, this phase-field parameter $u(\mathbf{x},t)$ has value "0" for liquid phase and "1" for the solid phase, and the phase transition layer containing the interface is intermediate value between 0 and 1 [97]. Governing equations for dendritic crystal growth are as follows [96, 97]:

$$\tau\frac{\partial u}{\partial t} = -\frac{\delta E(u)}{\delta u}, \tag{5.1}$$

$$\frac{\partial \hat{s}}{\partial t} = \nabla^2\hat{s} + k\frac{\partial u}{\partial t}. \tag{5.2}$$

The term $\frac{\delta E(u)}{\delta u}$ is the functional (variational) derivative of the square-gradient free energy functional $E(u)$ and δ signifies the functional (variational) derivative opera-

tor. The square-gradient free energy functional $E(u)$ given in equation (5.1) measures the energy of the system and in interface as well as distinguishes the solid and liquid phases. The negative sign in equation (5.1), shows that minimization of gradient free energy functional results the decline in total energy of the transition region. The square-gradient free energy functional for dendritic crystal growth is given by Wheeler *et. al.*. [99]:

$$E(u) = \int \left[F(u) + \frac{\varepsilon^2 (\nabla u)^2}{2} \right] d\mathbf{x}. \tag{5.3}$$

In equation (5.3), the double well free energy functional $F(u)$ is given by:

$$F(u) = \frac{u^4}{4} - \left(\frac{1}{2} - \frac{\hat{n}}{3} \right) u^3 + \left(\frac{1}{4} - \frac{\hat{n}}{2} \right) u^2, \tag{5.4}$$

which measures the growth of solid phase due to consumption of liquid phase, and gives two minimum values 1 and 0 for solid and liquid phases respectively as shown in figure 5.1 [100].

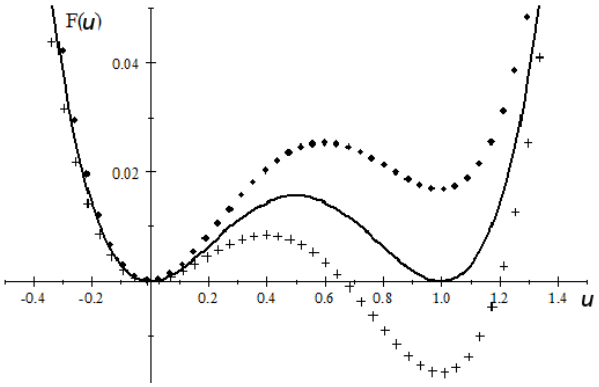


Figure 5.1: Ilustration of the free-energy functional: dotted line $(\hat{n} < 0)$, solid line $(\hat{n} = 0)$ and crossed line $(\hat{n} > 0)$.

The value of $\hat{n}$ is related difference between actual temperature $\hat{s}$ and melting temperature $\hat{s}_m$, and causes the growth of solid phase. The coefficients in equation (5.4) is simply chosen to force the two minima to be 0 and 1. One can adopt several prescriptions for $\hat{n}$ to fix the distance to equilibrium. However, we choose the Kobayashi's prescription $\hat{n}(\hat{s}) = \frac{\varpi}{\pi} \tan^{-1} [\zeta (\hat{s}_m - \hat{s})]$ where ϖ and ζ are material constants [22]. By taking $\varpi < 1$ ensures $|\hat{n}| < \frac{1}{2}$ for all values of $\hat{s}$. From figure 5.1, we can see that $\hat{n} > 0$ leads to the solid phase $(u = 1)$ whereas $\hat{n} < 0$ leads to the liquid phase $(u = 0)$ and $\hat{n} = 0$ is the phase equilibrium. The difference of two minimum values is proportional

to $\hat{n}$ (i.e. $F(0)-F(1)=\frac{\hat{n}}{6}$), which gives a difference of chemical potential of the both phases.

In the numerical description of dendrites growth, the anisotropy which exists on the solid/liquid interface must be considered. To take anisotropy into account, Kobayashi and Wheeler et al. [99], assumed an angular dependence of the interfacial width "ε" is function some angle ξ i.e. $\varepsilon=\varepsilon(\xi)$ is defined as:

$$\varepsilon=\varepsilon_0(1+\vartheta\cos(a_0(\xi-\xi_0))) \tag{5.5}$$

where ϕ is in the normal direction to the interface. In equation (5.5) a_0, ξ_0 and ϑ are modulation of interface, orientation of anisotropy and anisotropic mode number respectively. Since the thermodynamical driving forces enter the dendrite growth problem through the Gibbs-Thomson condition, which relates the local equilibrium temperature to the interface curvature. In multi-dimensions, interface curvature is a scalar quantity $\varepsilon=\varepsilon(\xi)$, where ξ is the angle between ∇u and a fixed reference axis. For the positive value of ϑ, interface curvature minima correspond to x-axis dendrite growth directions [98]. The growth of solid phase is due to thermal gradient in which the solid-liquid interface melts and reduces to the angle ξ. Equation (5.2) is the heat equation having source term $\kappa\frac{\partial u}{\partial t}$. The equations $(5.1-5.2)$ in two-dimensional form can be written as [99];

$$\tau\frac{\partial u}{\partial t} = \nabla.\left(\left(\varepsilon^2\mathbf{I}+\varepsilon\varepsilon'\mathbf{A}\right)\nabla u\right)+u(1-u)(u-0.5+\hat{n}), \tag{5.6}$$

$$\frac{\partial\hat{s}}{\partial t} = \nabla^2\hat{s}+\kappa\frac{\partial u}{\partial t}, \tag{5.7}$$

$$\text{with } \mathbf{I}=\begin{bmatrix}1&0\\0&1\end{bmatrix} \text{ and } \mathbf{A}=\begin{bmatrix}0&-1\\1&0\end{bmatrix},$$

where $\varepsilon'=\frac{d\varepsilon}{d\xi}$. The system of equations $(5.6-5.7)$ is subject to initial conditions

$$u_0(x,y)=\begin{cases}1, & (x^2-5)+(y^2-5)\leq 1\\ 0, & \text{otherwise.}\end{cases}$$

and

$$\hat{s}_0(x,y)=0,$$

and no-flux boundary conditions

$$\nabla u\cdot\nu=\nabla\hat{s}\cdot\nu=0. \tag{5.8}$$

5.3 Numerical Discretization

For the numerical solution of system of equations $(5.6-5.7)$, we have adapted the Q_1 conforming finite element method (FEM) [101] and fractional step θ-scheme [62] for spatial and temporal discretization.

5.3.1 Space Discretization

For spatial discretization in the FEM we consider the steady state of equations $(5.6-5.7)$.

$$-\nabla.\left(\left(\varepsilon^2\mathbf{I}+\varepsilon\varepsilon'\mathbf{A}\right)\nabla u\right)-u\left(1-u\right)\left(u-0.5+\hat{n}\right) = 0 \quad (5.9)$$

$$-\nabla^2\hat{s} = 0, \quad (5.10)$$

and imposing the no-flux boundary conditions (5.8).

The weak formulation of equations $(5.6-5.7)$ can be obtained by multiplying equations $(5.6-5.7)$ with appropriate test functions v and $\hat{w}$ respectively and applying integration over space domain. Find $u\in U$ and $\hat{s}\in\hat{S}$ such that $r_u(u,v)=0$ and $r_{\hat{s}}(\hat{s},\hat{w})=0 \quad \forall v\in V$ and $\forall\hat{w}\in\hat{W}$, with the continuous residuals having forms

$$r_u(u,v) = \int_\Omega \left(\varepsilon^2\mathbf{I}+\varepsilon\varepsilon'\mathbf{A}\right)\nabla u.\nabla v-\left(u\left(1-u\right)\left(u-0.5+\hat{n}\right)\right)vdx, \quad (5.11)$$

$$r_{\hat{s}}(\hat{s},\hat{w}) = \int_\Omega \nabla\hat{s}.\nabla\hat{w}dx. \quad (5.12)$$

5.3.2 Time Discretization

The system of equations $(5.6-5.7)$ is stiff system of PDEs. It is difficult to solve system of equations $(5.6-5.7)$ with some explicit methods due to their stability restrictions on time step size. So some implicit time stepping approaches is the better choice to over come the small time step stability restriction. For this purpose, we used the three-stage second-order diagonally implicit fractional-step θ-scheme [62]. For unsteady part of system of equation $(5.6-5.7)$, we find $u\in L_2(t_0,t_0+T;u_g+V(t))$ and $\hat{s}\in L_2(t_0,t_0+T;\hat{s}_g+\hat{W}(t))$

$$\frac{d}{dt}\int_\Omega uvdx+\int_\Omega\left(\varepsilon^2\mathbf{I}+\varepsilon\varepsilon'\mathbf{A}\right)\nabla u.\nabla v-\left(u\left(1-u\right)\left(u-0.5+\hat{n}\right)\right)vdx=0 \quad (5.13)$$

$$\forall v\in V(t), t\in\Sigma, \quad (5.14)$$

$$\frac{d}{dt}\int_\Omega\hat{s}\hat{w}dx-\frac{d}{dt}\int_\Omega\kappa u\hat{w}dx+\int_\Omega\nabla\hat{s}.\nabla\hat{w}dx=0, \quad (5.15)$$

$$\forall\hat{w}\in\hat{W}(t), t\in\Sigma, \quad (5.16)$$

where $V(t) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni u_g(t)|_{\Gamma_D} = g_u$, and $\hat{W}(t) = \{\hat{w} \in H^1(\Omega) : \hat{w} = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni \hat{s}_g(t)|_{\Gamma_D} = g_{\hat{s}}$. The system of equations $(5.13 - 5.15)$ can be written in a more compact way with residual forms as follows:

$$\frac{d}{dt} m^u(u,v) + r^u(u,v) = 0 \quad \forall v \in V(t), t \in \Sigma, \tag{5.17}$$

$$\frac{d}{dt} m^{\hat{s}}(\hat{s},\hat{w}) + r^{\hat{s}}(\hat{s},\hat{w}) = 0 \quad \forall \hat{w} \in \hat{W}(t), t \in \Sigma, \tag{5.18}$$

where $m^u(u,v) = \int_\Omega uv\,dx$ and $m^{\hat{s}}(\hat{s},\hat{w}) = \int_\Omega (\hat{s} - \kappa u)\hat{w}\,dx$ are the new temporal residual forms.

We choose conforming finite element spaces $V_h^{k,d}(\mathscr{T}_h,t)$ and $\hat{W}_h^{k,d}(\mathscr{T}_h,t)$, and the functions $u_{h,g_u}(t)$ and $\hat{s}_{h,g_{\hat{s}}}(t)$ depending on time so that $u_h(t) \in U_h(t) = u_{h,g_u}(t) + V_h(t)$ and $\hat{s}_h(t) \in \hat{S}_h(t) = \hat{s}_{h,g_{\hat{s}}}(t) + \hat{W}_h(t)$. So, the spatial and temporal residuals can be approximated as $r^u(u,v) \approx r_h^u(u_h,v_h)$ and $r^{\hat{s}}(\hat{s},\hat{w}) \approx r_h^{\hat{s}}(\hat{s}_h,\hat{w}_h)$, and $m^u(u,v) \approx m_h^u(u_h,v_h)$ and $m^{\hat{s}}(\hat{s},\hat{w}) \approx m_h^{\hat{s}}(\hat{s}_h,\hat{w}_h)$ respectively. Now for the integration of ODEs system, we subdivide the time interval into subintervals as follows

$$\overline{\Sigma} = \{t^0\} \cup (t^0,t^1] \cup \ldots \cup (t^{N-1},t^N]$$

with $t^0 = t_0$, $t^N = t_0 + T$, $t^{k-1} < t^k$ for $1 \le k \le N$ and set the time step as $\Delta t^k = t^{k+1} - t^k$. One of the simplest ODE integrators is the one-step-θ scheme that reads; find $u_h^{k+1} \in U_h(t^{k+1})$ and $\hat{s}_h^{k+1} \in \hat{S}_h(t^{k+1})$ such that

$$\frac{1}{\Delta t^k}(m_h^u(u_h^{k+1},v;t^{k+1}) - m_h^u(u_h^k,v;t^k)) + \theta r_h^u(u_h^{k+1},v;t^{k+1}) + (1-\theta) r_h^u(u_h^k,v;t^k) = 0 \quad \forall v \in V_h(t^{k+1}),$$

$$\frac{1}{\Delta t^k}(m_h^{\hat{s}}(\hat{s}_h^{k+1},\hat{w};t^{k+1}) - m_h^{\hat{s}}(\hat{s}_h^k,\hat{w};t^k)) + \theta r_h^{\hat{s}}(\hat{s}_h^{k+1},\hat{w};t^{k+1}) + (1-\theta) r_h^{\hat{s}}(\hat{s}_h^k,\hat{w};t^k) = 0 \quad \forall \hat{w} \in \hat{W}_h(t^{k+1}).$$

Reordering the terms for $(k+1)$ and k shows that the method give solution to a nonlinear system [61], i.e.,

$$m_h^u(u,v;t^{k+1}) + \Delta t^k \theta r_h^u(u,v;t^{k+1}) - m_h^u(u_h^k,v;t^k) + \Delta t^k (1-\theta) r_h^u(u_h^k,v;t^k) = 0 \quad \forall v \in V_h(t^{k+1}),$$

$$m_h^{\hat{s}}(\hat{s},\hat{w};t^{k+1}) + \Delta t^k \theta r_h^{\hat{s}}(\hat{s},\hat{w};t^{k+1}) - m_h^{\hat{s}}(\hat{s}_h^k,\hat{w};t^k) + \Delta t^k (1-\theta) r_h^{\hat{s}}(\hat{s}_h^k,\hat{w};t^k) = 0 \quad \forall \hat{w} \in \hat{W}_h(t^{k+1}).$$

At each time step, the resulting linear system is solved by Bi-Conjugate Gradient (BiCG) method while the nonlinear system is solved by Newton's method [54, 137]. To reduce the computational cost we applied adaptive time-stepping algorithm which is given as follows:

Algorithm 2 Adaptive time-stepping process

Given:$\mathbf{W_n} = \begin{bmatrix} u_n \\ \hat{s}_n \end{bmatrix}$, Tol=$10^{-6}$, $\Delta t_{\text{start}} = 10^{-4}$, $\Delta t_{\max} = 10^{-2}$, End Time $= 0.4$

while Time $<$ End Time **do**

Compute $\mathbf{W_{n+1}}$ using Δt

Calculate $e_{n+1} = ||\mathbf{W_{n+1}} - \mathbf{W_n}||$

if $e_{n+1} >$ Tol **then**

$\Delta t_n = \Delta t_n/2$

Recompute $\mathbf{W_{n+1}}$,

else

Update the time step.

If $\Delta t_n < \Delta t_{\max}$

$\Delta t = \min(\sigma \Delta t_n, \Delta t_{\max})$; where σ is adaptive factor.

end if

end while

5.4 Results and Discussions

In this section, we assess the time step stability of the proposed DIFST scheme. Our goal is to find the maximum time step size at which a given dendritic crystal growth model is solvable. We also provide efficiency and robustness of the DIFST scheme in comparison with DIRK2 and DIRK3 schemes [61] using different values of adaptivity factor in Algorithm 1. Table 4.1 shows that both DIFST and DIRK3 schemes work for small adaptive factor $\sigma = 1.1$ but DIRK2 scheme fails after 46 time steps. Tables 4.2 and 4.3 show that for $\sigma = 1.2$ & 2.0 both DIRK2 and DIRK3 schemes fail after few time steps while DIFST scheme works well resulting in the reduction of simulation time. We have calculated the computational cost in term of maximum linear iterations (MAX L-IT), maximum nonlinear iterations (MAX NL-IT), central processing unit time (CPU

TIME) and linear solver time (LS TIME).

Schemes	TIME STEPS	MAX L-IT	MAX NL-IT	CPU TIME	LS TIME
DIFST	79	1213	538	2474.7	086.07
DIRK3	79	3370	603	2903.9	259.58
DIRK2	Fails after 46 time steps with $\Delta t > 8.7017 \times 10^{-4}$				

Table 5.1: For adaptive factor $\sigma = 1.1$

Schemes	TIME STEPS	MAX L-IT	MAX NL-IT	CPU TIME	LS TIME
DIFST	61	1174	471	2254	85.99
DIRK3	Fails after 27 time steps with $\Delta t > 9.5396 \times 10^{-3}$				
DIRK2	Fails after 25 time steps with $\Delta t > 1.0 \times 10^{-2}$				

Table 5.2: For adaptive factor $\sigma = 1.2$

Schemes	TIME STEPS	MAX L-IT	MAX NL-IT	CPU TIME	LS TIME
DIFST	46	1116	403	1850.1	77.555
DIRK3	Fails after 6 time steps with $\Delta t > 1.0 \times 10^{-2}$				
DIRK2	Fails after 9 steps with $\Delta t > 4.6 \times 10^{-3}$				

Table 5.3: For adaptive factor $\sigma = 2.0$

Since in two-phase problems, the coupling of interface equation plays a crucial role and limits the stability and range of small time steps due to moving boundary. The two-phase flow models like dendritic crystal growth, this restriction can increase computational cost. Even in those cases when the interface is almost stationary, large time steps will lead to oscillations and finally destruction of the interface. To handle this issue we proposed technique that could lift the above time step restriction. We demonstrate performance of present numerical scheme by computing the error with constant time step and adaptive time steps by using the following formula:

$$Error = \mid u^{k+1} - u^{k} \mid .$$

Figure 5.2(a) shows local temporal error (magnitude of difference between two consecutive time steps) for constant time step $\Delta t = 10^{-4}$. In zoom view of figure 5.2(a),

we can observe that with small time steps the scheme becomes stable soon. Figure 5.2$(b-d)$ shows the errors with adaptive time steps by using Algorithm 1 with different values of adaptivity factor given in Tables 4.1-4.3 and shows that DIFST scheme is more stable for large time steps.

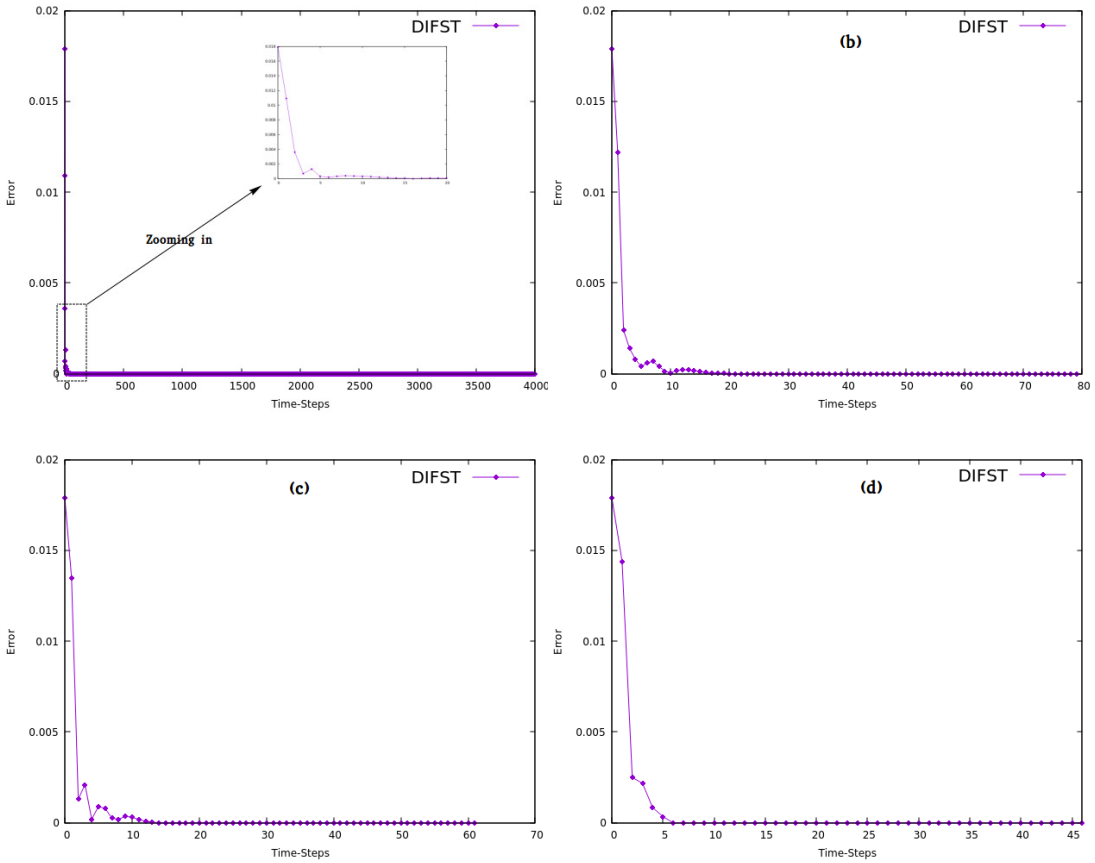


Figure 5.2: Error with constant and adaptive time steps

Figure 5.3 shows the comparison of error between DIFST and DIRK3 schemes, this shows that the present DIFST scheme is more stable as compare to DIRK3 scheme.
The computed results for dendritic growth u and temperature $\hat{s}$ at different time levels using Algorithm 1 are given in figure 5.4 and compared with method provided by T. Biben [97] as shown in figure 5.5. From qualitative comparison between figures 5.4 and 5.5, we observed faster growth, better resolution with pattern formation using the proposed method.

5.5 Parametric Study

In this section, we simulated the system of equations $(5.6-5.7)$ at $t = 0.2$ and investigated the effects of k, $\hat{s}_m$, τ, ε_0 and ϑ on shape and growth of dendritic. Here,

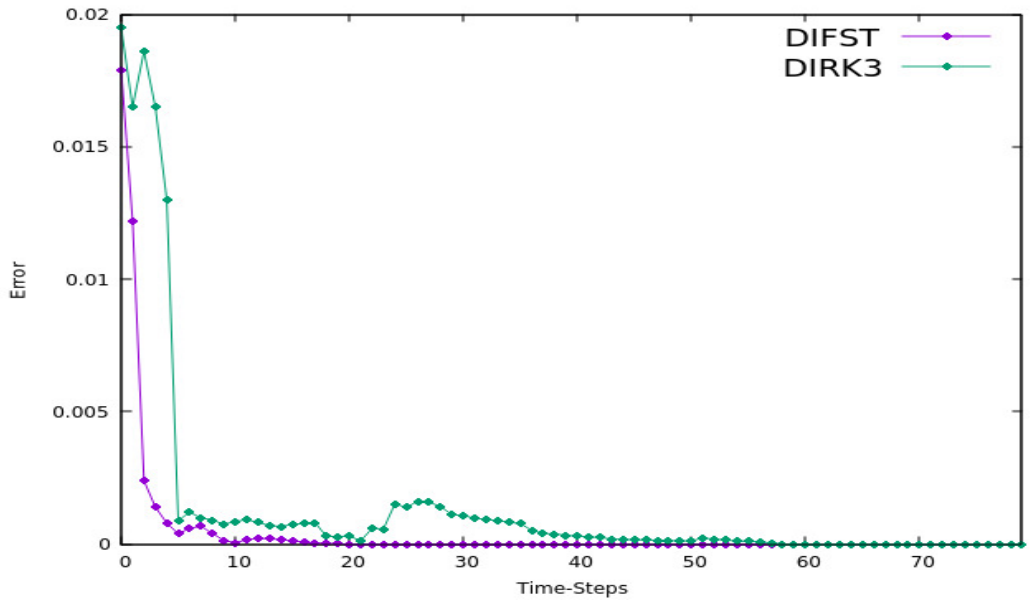


Figure 5.3: Comparison of error between DIFST and DIRK3 schemes.

we have fixed values of $\varpi = 0.9$, $\zeta = 10.0$, $a_0 = 6.0$ and $\theta_0 = 1.57$ with mesh size $h = \Delta x = \Delta y = 0.03$ over domain $[0,300] \times [0,300]$.

5.5.1 Effect of Latent Heat

In the solidification process, the heat effects play the significant role. So, it is important to study the effects of latent heat "k" in dendritic crystal growth phenomenon. Figure 5.6 shows that large values of "k" account for the eviction of more heat from the interfacial region. As latent heat increases from 0.5 to 2.0, the tip radius and velocity of side branches also increases [102]. Small value of "k" can inhibit the growth of side branches.

5.5.2 Effect of Melting Temperature

Simulation results for melting temperature $\hat{s}_m$ at $t = 0.2$ are given in figure 5.7 to shows its effect on the growth, size and shape of dendrite. It is observed that increasing the value of $\hat{s}_m$ accelerates growth process. It can also be seen from figure 5.7 that increasing $\hat{s}_m$ increases the tip velocity, while tip radius of side branches has an inverse effect. By increasing $\hat{s}_m$, the growth rate decreases i.e. the dendrite have larger radii while side branches grow with less speed. It is to be noted that increasing $\hat{s}_m$ beyond the melting point will melt the structure.

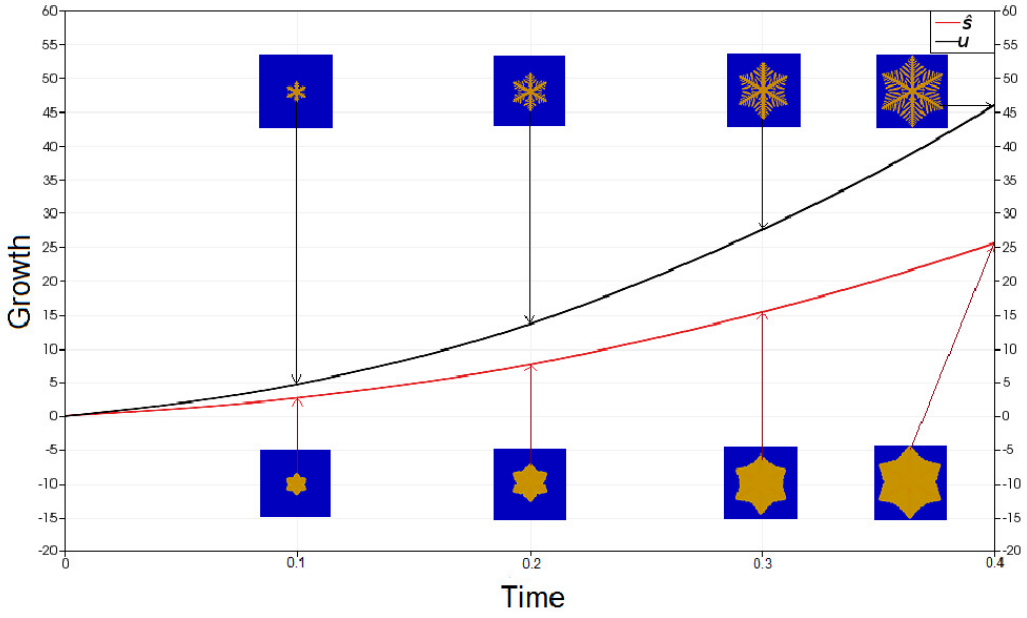


Figure 5.4: Calculation of u and $\hat{s}$ using proposed method.

5.5.3 Effect of Relaxation Time

The relaxation time measurement can provide many insights into molecular and atomic structures along with rates and mechanisms of chemical reactions. In figure 5.8 , the relaxation time effect on dendrite for $\tau = 0.0002$, 0.0003, 0.0004 and 0.0005 is given at time$t = 0.2$. It shows that for $\tau = 0.0002$, the dendrite grow faster than other values which shows that for small values of τ, the atoms need less time to attains their equilibrium state, while for large values of τ it reaches their equilibrium state lately. From figure 5.8, we notice that increasing from $\tau = 0.0002$ to $\tau = 0.0005$, the tip radius and velocity of side branches decrease.

5.5.4 Effect of Interfacial Width

dendritic growth is is free boundary problem base one phase-field models, so the interfacial width plays significant role on shape and speed of side branches of dendriticies. Figure 5.9 shows computed results for $\varepsilon_0 = 0.009, 0.01, 0.011$ and 0.012, and observed that growth process is faster for larger values of ε_0.

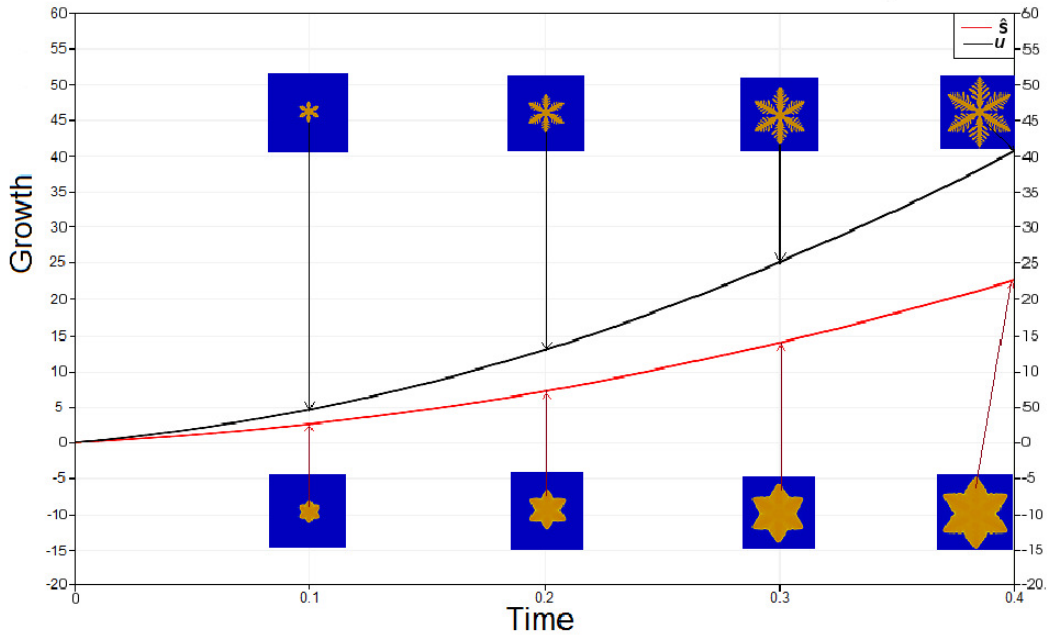


Figure 5.5: Calculation of u and $\hat{s}$ using explicit Euler's method.

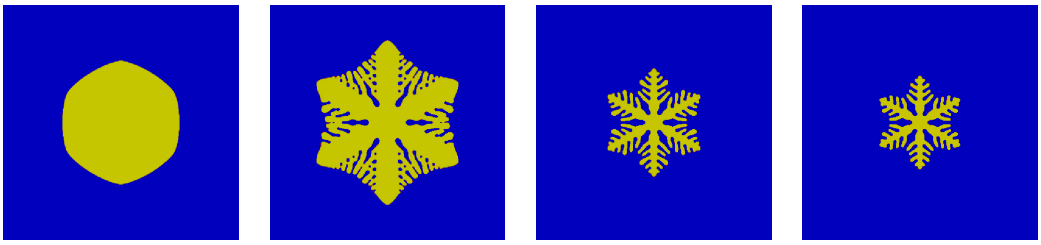


Figure 5.6: Effect of latent heat $k = 0.5, 1.0, 1.8$ and 2.0 (left to right) on dendrite.

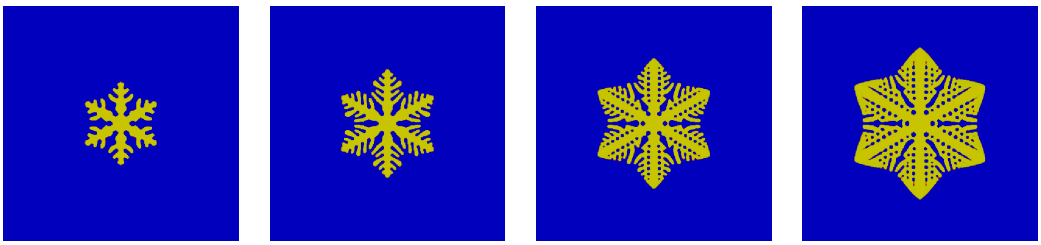


Figure 5.7: Effect of melting temperature $\hat{s}_m = 0.9, 1.0, 1.2$ and 1.3 (left to right) on dendrite.

5.6 Summary

In this chapter, we proposed the efficient time adaptive algorithm for the numerical simulation of nonlinear system of PDE's for dendritic crystal growth. The space discretization is done by Q_1 conforming FEM and diagonally implicit fractional step θ (DIFST) scheme to march in time. The DIFST scheme is compared with traditional diagonally implicit Runge-Kutta (DIRK) scheme of order 2 and 3 for small by using

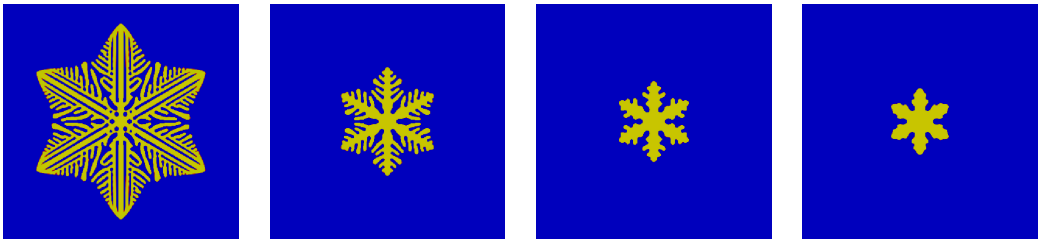


Figure 5.8: Effect of relaxation time $\tau = 0.0002, 0.0003, 0.0004$ and 0.0005 (left to right) on dendrite.

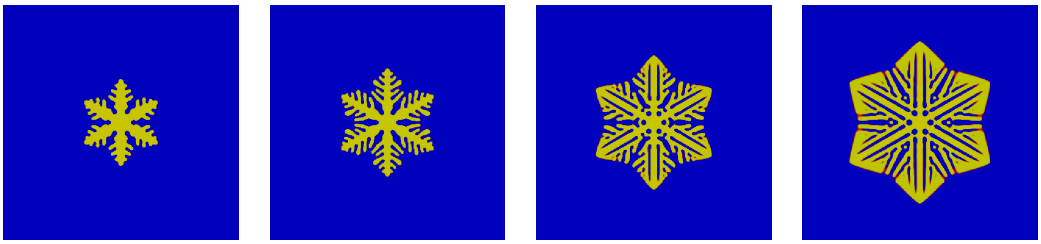


Figure 5.9: Effect of interfacial width $\varepsilon_0 = 0.009, 0.01, 0.011$ and 0.012 (left to right) on dendrite.

large adaptive factors. Efficiency and robustness of proposed scheme is observed for numerical simulation of dendritic crystal growth. We concluded that due to the better computational performance, the proposed DIFST scheme may allow to choose small interfacial width for dendritic crystal growth model, which may come more closer to physics of the problem.

Chapter 6

Mathematical Modeling and Numerical Simulation of Tumor Hypoxia

This chapter deals with the mathematical formulation of tumor hypoxia-targeting and its numerical solution. The well-posedness of the governing partial differential equations is provided. For the purpose of numerical simulations, the conforming Q_1 FEM for space discretization and second-order diagonally implicit fractional step θ-scheme for temporal discretization, proposed in chapter 2 are used.

6.1 Introduction

Solid tumor include areas where oxygen concentration is very low called hypoxia, often in the surrounding areas of necrosis. Hypoxic cells in these areas are resistant to chemotherapy and radiation therapy. The presence of hypoxia and necrosis enables tumor-selective treatment, including hypoxia-activated prodrugs, tumor hypoxia-specific gene therapy and tumor-targeting bacterial therapy. Cancer or rapid creation of abnormal/tumor cells that grow beyond their usual boundaries are among the leading causes of morbidity and mortality worldwide [106]. Therefore, advanced countries have allocated a large amount of research funding for the development of new methods and drugs to treat cancer. Due to availability of early identification tools and prodrugs techniques, the chances of successful cancer treatment has increased [107]. The formation of capillaries and new blood vessels from the existing vasculogenic network is referred to as angiogenesis [108]. The link between angiogenesis and tumor growth was first postulated in 1971 [110] and subsequently numerous studies have demonstrated that without the process of angiogenesis, growth of a tumor is restricted due to a limitation of oxygen and other nutrients derived from the blood supply [111, 112]. Since tumor cells require proper blood supply, provision of oxygen and other nutrients, their growth is limited to a certain size. Therefore, in order to grow beyond a certain size, tumors need to induce an angiogenic response from the blood vessels proximal to the tumor micro-environment [113, 114]. Hypoxic conditions (transient hypoxic stress) in the core of the developing tumor and the resultant expression of hypoxia inducible factors (HIFs) in tumor cells are potent mediators of angiogenesis [115] which is required for blood and nutrients supply to proliferating cells. As the proliferating tumor cells consume most of available oxygen in the vicinity of normal cells, this decrease a oxygen tension may be harmful for the survival of later. However, many other cells can survive even under poorly oxygenated (hypoxic) conditions [107].

It is well-established that hypoxic tumor cells are resistant to radiation therapy [116, 117], but the clinical impact of hypoxic tumor cells extends beyond the treatment of localized primary tumors with ionizing radiation. Hypoxic tumor cells promote tumor progression and metastasis through a variety of direct and indirect mechanisms, and hypoxic tumor cells, therefore, represent a significant impediment to successful cancer therapy [116, 117, 118]. Therefore, the study of hypoxic tumor cells has significant impact in cancer treatment [119].

In comparison to molecular biology, cell biology and anti-cancer drugs research, the contribution of mathematics for cancer treatment is relatively low [119, 120, 121]. A mathematical model of tumor growth is a mathematical expression for the growth for abnormal cells over time [122]. Estimating the number of cancer cells in a tumor and tracking of cells migration and position as a function of time is a challenging task both in applied sciences and cancer biology. Although, different mathematical strategies are developed for tumor growth and cancer treatment. For example, in Ref. [123], several simple differential equations are illustrated to describe the tumor growth and anti-tumor treatments. Particularly, in the case of an invasive tumor growth, the kinetics of nutrients concentration and cell density is given by a system of reaction-diffusion equations [124]. Some qualitative analysis and numerical simulations of the cancer models are given in [125, 126, 127]. Particularly in Ref.[127], A.V. Kolobov et. al., have proposed a mathematical model for the study of invasive tumor growth and have provided a traveling wave solution. The aim of this work is to extend the work done by A.V. Kolobov by considering the oxygen decay parameter to target tumor hypoxia in cancer therapy.

6.2 Mathematical Model

Tumors are regarded as a colony of live and dead cells surrounded by normal tissues. The live tumor cells possess mobility, divide at a constant rate, and begin to die due to lack of nutrients. Although cells division and mobility require a large variety of nutrients, red model assumes only the lack of oxygen (i.e. nutrients) to determines the death of malignant cells [127]. It also assumes that oxygen consumption by normal non-proliferating cells is negligibly small when compared with actively dividing tumor cells [128]. Since tumor cells are self-sufficient in growth signals and divide more rapidly

than normal cells which is why their oxygen consumption is comparatively more than the non-proliferating (normal) cells [129, 130]. The greater the number of cells, the more will be their requirement for oxygen which is vital for metabolism, growth and division of cells. Further assumptions are considered that the surrounding normal tissue does not hinder tumor cell movement and proliferation. Therefore, the model given in [127] is based on the live cells with density "u" which divide at constant rate β and randomly move (diffuse) with constant coefficient D_u, die at low oxygen concentration "s" at a rate of cell death $P(s)$. Since oxygen is essential for the growth of the tumor cells, its distribution is determined by the diffusion constant coefficient D_s and nutrient (oxygen) consumption rate by live tumor cells, factor of μ.

Decay of oxygen is natural factor and it causes tumor hypoxia. However, numerous studies have shown that tumor hypoxia, in which portions of the tumor have significantly low oxygen concentrations, is in fact linked with more aggressive tumor behavior and poorer prognosis [131, 132]. So the decay of oxygen is considered as a factor of ρ in the model proposed by Kolobov et al. [127], which has significant effects in cancer growth. The interactions of tumor cells and the oxygen supply is described by a system of following partial differential equations;

$$\frac{\partial u}{\partial t} = \underbrace{D_u \nabla^2 u}_{\text{diffusion}} - \underbrace{P(s)\,u}_{\text{death rate}} + \underbrace{\beta u}_{\text{cells divission}}, \tag{6.1}$$

$$\frac{\partial s}{\partial t} = \underbrace{D_s \nabla^2 s}_{\text{diffusion}} - \underbrace{\mu u}_{\text{absorption}} - \underbrace{\rho s}_{\text{oxygen decay}}, \tag{6.2}$$

subject to initial conditions

$$u(x,0) = u_0$$

$$s(x,0) = s_0,$$

and boundary conditions

$$(u, \nabla s) \to (0,0) \text{ when } x \to -\infty,$$

$$(u, s) \to (0,1) \text{ when } x \to +\infty.$$

In Eq. (6.1), $P(s) = \frac{\lambda}{2}[1 - \tanh(\frac{s - s_{crit}}{\varepsilon})]$ is death cell rate and, λ, ε and s_{crit} are positive constants describing, maximal death rate, characteristic deviation of s from s_{crit} and critical nutrient (oxygen) concentration respectively [127]. Equation $(6.1 - 6.2)$ have written in dimensionless form taking the time and length scales $T_0 = 10^6 second$ and $L =$

$5 \times 10^{-2}\ cm$, cells density scale $c_{max} = 10^7\ \ cells/cm^3$, and oxygen concentration scale $s_{max} = 10^{-7}\ \ mol/cm^3$ (oxygen concentration near blood vessels). The dimensionless proliferation rate corresponds in the order of magnitude to one division in 100 *hours* while for cell the diffusion coefficient is $D_u = 2.5 \times 10^{-9}\ cm^2/second$ and for oxygen the diffusion coefficient is $D_s = 2.5 \times 10^{-5}\ cm^2/second$.

6.3 Mathematical Analysis

In this section, we discuss well-posedness of the problem by considering Eq. (6.1) and Eq. (6.2) in compact form as follows;

$$\frac{\partial}{\partial t}\begin{bmatrix} \mathrm{u} \\ \mathrm{s} \end{bmatrix} = \begin{bmatrix} \nabla^2 + \beta & 0 \\ 0 & \nabla^2 - \rho \end{bmatrix}\begin{bmatrix} \mathrm{u} \\ \mathrm{s} \end{bmatrix} + \begin{bmatrix} -P(s)\,u \\ -\mu u \end{bmatrix} \tag{6.3}$$

subject to initial conditions

$$u(x,t=0) = u_0,\ \ s(x,t=0) = s_0,\ \forall x \in \mathbb{R} \tag{6.4}$$

and boundary conditions

$$(u, \nabla s) \to (0,0)\ \text{ when } x \to -\infty,\ \ (u,s) \to (0,1)\ \text{ when } x \to +\infty. \tag{6.5}$$

For $1 \leq p < \infty$, we define the Banach space Φ_p such that
$\Phi_p = \{(u,s) = \psi : \mathbb{R} \longrightarrow \mathbb{R},\ \psi \text{ is bounded and uniformly continuous and } \psi \in L^p(\mathbb{R})\}$ induced with norm
$\|\psi\|_p = \sup_{\xi \in \mathbb{R}} |\psi(\xi)| + \left(\int_{-\infty}^{+\infty} |\psi(\xi)|^p d\xi \right)^{\frac{1}{p}}$. For $p \geq q$ and $q \in \mathbb{N}$, we have a continuous inclusion $W^{1,q}(\mathbb{R}) \subset \Phi_p$ [134]. Let $n \in \mathbb{N}$ and consider the Banach space $\Phi_p^{(n)} = \left\{ \psi \in \Phi_p : \psi', \psi'', ..., \psi^n \in L^p(\mathbb{R}) \right\}$ with the norm $\|\psi\|_p^n = \sum_{j=0}^{n} \|\psi^{(j)}\|_p,\ \psi \in \Phi_p^{(n)}$ and using the results given in [133] (p-21,22), the system of equation (6.3) can be written in abstract parabolic evolution equation as follows:

$$\frac{d\psi}{dt} = A\psi + F(\psi), \quad \psi(0) = \psi_0, \tag{6.6}$$

with $F : \Phi \longrightarrow \Phi$ is a nonlinear function given by $F(\psi) = (-P(s)\,u, -\mu u)$ and $\psi_0 =$

(u_0, s_0) with

$$D(A) = \left\{ (u,s) \in \Phi : (u,s) \in \Phi_p^{(2)}, \begin{array}{c} (u, \nabla s) = (0,0), \text{ as } x \to -\infty, \\ (u,s) = (0,1), \text{as } x \to +\infty \end{array} \right\},$$

and

$$A\psi = \left(D_u \nabla^2 u + \beta u,\ D_s \nabla^2 s - \rho s \right).$$

Now, using variation of constant formula [134] and the result given in [135] (p-105), solution of Eq. (6.6) is given by

$$\psi(t) = \psi_0 e^{At} + \int_0^t e^{A(t-\xi)} F(\psi(\xi))\, d\xi, \ \ t \geq 0, \tag{6.7}$$

where

$$\begin{aligned} e^{At} &= \begin{pmatrix} T_1(t) & 0 \\ 0 & T_2(t) \end{pmatrix} \\ (T_1(t) u_0)(x) &= e^{\gamma t} \int_{\mathbf{R}} \Gamma_1(t, x-y)\, u_0(y)\, dy, \\ (T_2(t) s_0)(x) &= e^{-\rho t} \int_{\mathbf{R}} \Gamma_2(t, x-y)\, s_0(y)\, dy. \end{aligned}$$

Γ_1 and Γ_2 are the heat kernel functions associated with equations $u_t = D_u \nabla^2 u$ and $s_t = D_s \nabla^2 s$ respectively. Note that the equations $u_t = D_u \nabla^2 u + \beta u$ and $s_t = D_s \nabla^2 s - \rho s$ generate the linear semi groups $T_1(t)$ and $T_2(t)$ respectively. Thus for $t \geq 0$, Eq. (6.7) take the form

$$u(.,t,\psi_0) = T_1(t) u_0 - \int_0^t T_1(t-\xi) P(s(\xi)) u(\xi)\, d\xi, \tag{6.8}$$

$$s(.,t,\psi_0) = T_2(t) s_0 - \mu \int_0^t T_2(t-\xi) u(\xi)\, d\xi. \tag{6.9}$$

Definition 6.1 *We say that the function $\psi : [0, +\infty) \to \Phi$ is a mild solution of Eq. (6.6) if $\psi \in C([0,+\infty), \Phi)$ and satisfies Eq. (6.7).*

We have the following result on existence and uniqueness of equation (6.3) and (6.5).

Theorem 6.1 *[135] Assume that $F : [0,+\infty) \times \Phi \to \Phi$ be continuous function in "t" and uniformly Lipschitz continuous on Φ. If A is generator of strongly continuous semigroup $(T(t))_{t \geq 0}$, then for any $\psi_0 = (u_0, s_0) \in \Phi$ the system (6.3) with (6.5) has unique mild solution $\psi(x,t) = (u(x,t,\psi_0), s(x,t,\psi_0))$ for all $t \geq 0$ with $\psi(.,t,\psi_0) = \psi_0$.*

Proof 1 *Under above assumptions, definition of P and* $F(\psi) = (-P(s)u, -\mu u)$*, and consequence of the abstract results of [135](p-185-190), there exist a unique mild solution for system* (6.3).

Now, define the family of operators $\{T(t) : t \geq 0\}$ on Φ by

$$\begin{aligned}(T(t)\omega)(x) &= e^{tA}\omega(x), \\ &= \psi(x,t,\omega) \\ &= (u(x,t,\omega), s(x,t,\omega)), \quad \forall x \in \mathbb{R}, \quad t \geq 0.\end{aligned}$$

For any $(t_0, \omega_0) \in [0,\infty) \times \Phi$, we have

$$\|T(t)(\omega) - T(t_0)\omega_0\| \leq \|T(t)(\omega) - T(t)\omega_0\| + \|T(t)(\omega_0) - T(t_0)\omega_0\|. \quad (6.10)$$

From the inequality (6.10), it follows that $T(t)\omega$ is continuous at (t_0, ω_0), and thus the solution $\psi(x,t,\omega)$ of equation (6.3) depends continuously on initial data.

6.4 Numerical Discretization

For the numerical solution of system of equations $(6.1) - (6.2)$, we have adapted the Q_1 conforming FEM [101] for spatial discretization and fractional step θ-scheme (DIF-STS) [62] for temporal discretization. For spatial discretization in the FEM we consider the steady state of equations $(6.1) - (6.2)$.

$$-D_u\nabla^2 u + P(s)u - \beta u, = 0 \quad (6.11)$$

$$-D_s\nabla^2 s + \mu u + \rho s = 0, \quad (6.12)$$

and imposing the no-flux boundary conditions

$$-\nabla u.\nu = -\nabla s.\nu = 0 \quad (6.13)$$

The weak formulation of equations $(6.1) - (6.2)$ can be obtained by multiplying equations $(6.1) - (6.2)$ with appropriate test functions v and w respectively, then applying integration by parts to evaluate the integrals. Find $u \in U$ and $s \in S$ such that $r_u(u,v) = 0$ and $r_s(s,w) = 0 \quad \forall v \in V$ and $\forall w \in W$, with the continuous residuals having

forms

$$r_u(u,v) = \int_\Omega D_u \nabla u.\nabla v + (P(s)u - \beta u)v dx, \tag{6.14}$$

$$r_s(s,w) = \int_\Omega D_s \nabla s.\nabla w + (u + \rho s)w dx. \tag{6.15}$$

For time discretization, we have adapted the second-order three-stage diagonally implicit fractional-step θ-scheme [62]. For time dependent system of equations $(6.1) - (6.2)$, we find $u \in L_2(t_0, t_0+T; u_g + V(t))$ and $s \in L_2(t_0, t_0+T; s_g + W(t))$

$$\frac{d}{dt}\int_\Omega uv\,dx + \int_\Omega D_u \nabla u.\nabla v + (P(s)u - \beta u)v dx = 0, \qquad \forall v \in V(t), \text{ and } t \in \Sigma, \tag{6.16}$$

$$\frac{d}{dt}\int_\Omega uv\,dx + \int_\Omega D_s \nabla s.\nabla w + (u + \rho s)w dx. = 0, \qquad \forall w \in W(t), \text{ and } t \in \Sigma. \tag{6.17}$$

where $V(t) = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni u_g(t)|_{\Gamma_D} = g_u$, and $W(t) = \{w \in H^1(\Omega) : w = 0 \text{ on } \Gamma_D(t)\}$ and $H^1(\Omega) \ni s_g(t)|_{\Gamma_D} = g_s$ respectively. Here $H^1(\Omega)$ is Hilbert space. The equations $(6.16) - (6.17)$ can be written in a more compact way with residual forms as follows:

$$\frac{d}{dt}m^u(u,v) + r^u(u,v) = 0 \quad \forall v \in V(t), t \in \Sigma, \tag{6.18}$$

$$\frac{d}{dt}m^s(s,w) + r^s(s,w) = 0 \quad \forall w \in W(t), t \in \Sigma, \tag{6.19}$$

where $m^u(u,v) = \int_\Omega uv\,dx$ and $m^s(s,w) = \int_\Omega sw\,dx$ are the new temporal residual forms. We choose conforming finite element spaces $V_h^{k,d}(\mathscr{T}_h, t)$ and $W_h^{k,d}(\mathscr{T}_h, t)$, and the functions $u_{h,g_u}(t)$ and $s_{h,g_s}(t)$ depending on time so that $u_h(t) \in U_h(t) = u_{h,g_u}(t) + V_h(t)$ and $s_h(t) \in S_h(t) = s_{h,g_s}(t) + W_h(t)$. So, the spatial and temporal residuals can be approximated as $r^u(u,v) \approx r_h^u(u_h, v_h)$ and $r^s(s,w) \approx r_h^s(s_h, w_h)$, and $m^u(u,v) \approx m_h^u(u_h, v_h)$ and $m^s(s,w) \approx m_h^s(s_h, w_h)$ respectively. Now for the integration of ODEs system, we subdivide the time interval into subintervals as follows

$$\overline{\Sigma} = \{t^0\} \cup (t^0, t^1] \cup \ldots \cup (t^{N-1}, t^N]$$

with $t^0 = t_0$, $t^N = t_0 + T$, $t^{k-1} < t^k$ for $1 \le k \le N$ and set the time step as $\Delta t^k = t^{k+1} - t^k$. One of the simplest ODE integrators is the one-step-θ scheme that reads; find $u_h^{k+1} \in U_h(t^{k+1})$ and $s_h^{k+1} \in S_h(t^{k+1})$ such that

$$\frac{1}{\Delta t^k}(m_h^u(u_h^{k+1}, v; t^{k+1}) - m_h^u(u_h^k, v; t^k)) + \theta r_h^u(u_h^{k+1}, v; t^{k+1}) + (1-\theta) r_h^u(u_h^k, v; t^k) = 0$$
$$\forall v \in V_h(t^{k+1}),$$

$$\frac{1}{\Delta t^k}(m_h^s(s_h^{k+1}, w; t^{k+1}) - m_h^s(s_h^k, w; t^k)) + \theta r_h^s(s_h^{k+1}, w; t^{k+1}) + (1-\theta) r_h^s(s_h^k, w; t^k) = 0$$
$$\forall w \in W_h(t^{k+1}).$$

Reordering the terms for $(k+1)$ and k shows that the method give solution to a nonlinear system [61], i.e.,

$$m_h^u(u,v;t^{k+1}) + \Delta t^k \theta r_h^u(u,v;t^{k+1}) - m_h^u(u_h^k,v;t^k) + \Delta t^k(1-\theta) r_h^u(u_h^k,v;t^k) = 0$$
$$\forall v \in V_h(t^{k+1}),$$
$$m_h^s(s,w;t^{k+1}) + \Delta t^k \theta r_h^s(s,w;t^{k+1}) - m_h^s(s_h^k,w;t^k) + \Delta t^k(1-\theta) r_h^s(s_h^k,w;t^k) = 0$$
$$\forall w \in W_h(t^{k+1}).$$

At each time step, the resulting linear system is solved by Bi-Conjugate Gradient (BiCG) method while the nonlinear system is solved by Newton's method [137].

6.5 Results and Discussion

In this section, we provide computed results by solving model for tumor hypoxia. The initial conditions "u_0" for density of live tumor cells is $u_0 = 0.1e^{-0.0025\mathbf{x}^2}$ and "s_0" for oxygen concentration is $s_0 = 1$ with no-flux boundary conditions over domain $\Omega = [-1000, 1000]$. In all computations, the other parameters are fixed as $D_s = 10^4$, $D_u = 10$, $\lambda = 0.2$, $\varepsilon = 0.01$, $s_{crit} = 0.3$, $\beta = 0.1$ and $\mu = 1$.

6.5.1 Time Accuracy

In this subsection, we have computed the order of time accuracy as well as number of iterations for linear and nonlinear solvers. The order of accuracy is calculated using the formula

$$O(\Delta t) = \frac{\log\left(\frac{\psi_{2\Delta t} - \psi_{4\Delta t}}{\psi_{\Delta t} - \psi_{2\Delta t}}\right)}{\log 2},$$

where $\psi = (u,s)$. Results are obtained for $\Delta t = 0.4, 0.2, 0.1, 0.05, 0.25$ and 0.0125 for $t = 26$ in 1D case using a fixed tolerance of 10^{-6} for BiCG linear solver. Table 6.1 shows the number of iteration for nonlinear and linear solvers while from Table 6.2, it is clear that the time discretization scheme is almost 2nd-order accurate. We have used $\Delta t = 0.05$ for all the simulations.

6.5.2 Results and Discussions

First, we validate our method using the decay factor $\rho = 0.0$ by comparing computed results with the reference solution [107]. The reference solution is obtained using plot

Table 6.1: Number of iterations for nonlinear and linear solvers

Δt	Time steps	nonlinear iterations	linear iterations
0.4	65	201	1217
0.2	130	400	2551
0.1	260	797	3193
0.05	520	1589	4717
0.025	1040	3164	6305
0.0125	2080	6303	12543

Table 6.2: Order of accuracy for time discretization scheme

Δt	c	$O(\Delta t)$	s	$O(\Delta t)$
0.4	0.7071789	...	0.3106530	...
0.2	0.7074093	...	0.3106663	...
0.1	0.7074710	1.9008	0.3106697	1.9678
0.05	0.7074857	2.0695	0.3106705	2.0875
0.025	0.7074893	2.0297	0.3106707	2.0
0.0125	0.7074902	2.0	0.31067075	2.0

digitizer software which is a Java program used to digitize scanned plots of functional data [138]. At a fixed $\mathbf{x} = 0$, figure 6.1 (left) shows that initially the growth of tumor cells is aggressive which reaches to its maximum value at time $t \approx 26.65$ of about 317 days 13 hours and 12 minutes, then start decaying till it attains the steady state at time $t \approx 200$ of about 6.5 years. We see that the jump discontinuity of c near $t \approx 26.65$ is due to the aggressive growth of tumor cells when maximum oxygen is available. However, when oxygen concentration declines significantly the cells begin to die. Figure 6.1(right) shows the asymptotic behavior from $t \approx 40$ to $t \approx 50$ for $\rho = 0.00$. By increasing the value of decaying parameter, the asymptotic behavior in growth of live tumor cells decreases and vanishes with $\rho = 0.01$. Table 6.3 shows the maximum age of live cells where it reaches its maximum value using different values of decay parameter ρ. We observed that the age of growing tumor cells decreases with the increase of ρ.

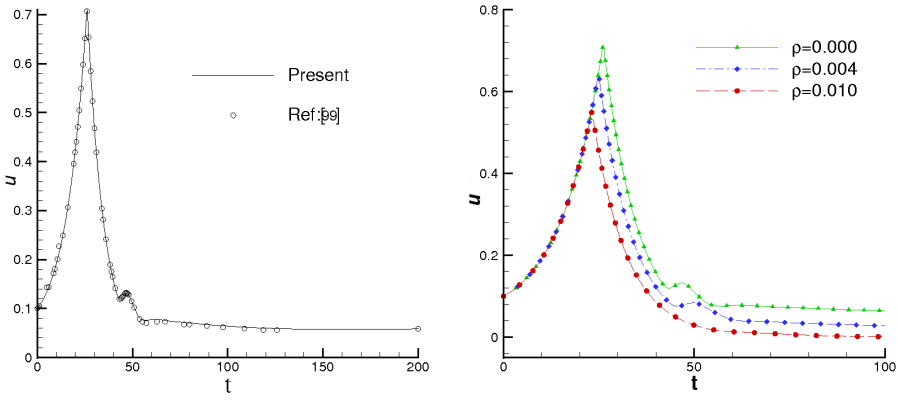


Figure 6.1: Comparison with reference result (left) and effect of ρ on asymptotic behavior of live tumor cells density "u" (right).

Table 6.3: Maximum age of growing tumor cells

ρ	Age
0.000	317 days 13 hours 12 mints
0.002	310 days 6 hours
0.004	304 days 9 hours 12 mints
0.006	283 days 11 hours 36 mints
0.008	275 days 13 hours 12 mints
0.010	267 days 16 hours

Further, we studied the effects of cells division rate β, maximal death rate λ and nutrient (oxygen) consumption rate by live tumor cells μ for $\rho = 0.004$. Figure 6.2 (left) shows that by increasing the value of $\beta = 0.0$ to 0.15, the tumor cells density is increasing. It is observed that for $\beta \leq 0.01$, number of live tumor cells will not increase. Also, from figure 6.2 (right), it is shown that the value of β directly effects on the oxygen concentration ("s") and observed that the value of β should be less than 0.15 to keep the oxygen concentration positive. By increasing the value of λ, the faster convergence of live tumor cells density to its steady-state and oxygen to its critical value $s_{cri} = 0.3$ are observed as shown in figure 6.3 (left) and 6.3 (right) respectively. Figure 6.4 (left) and 6.4 (right) shows the effects of oxygen consumption rate μ on the density of live tumor cells and oxygen concentration. It can be seen from figure 6.4 (left), as we increase the value of μ the tumor cells density is decreasing. It also can be observe form figure 6.4

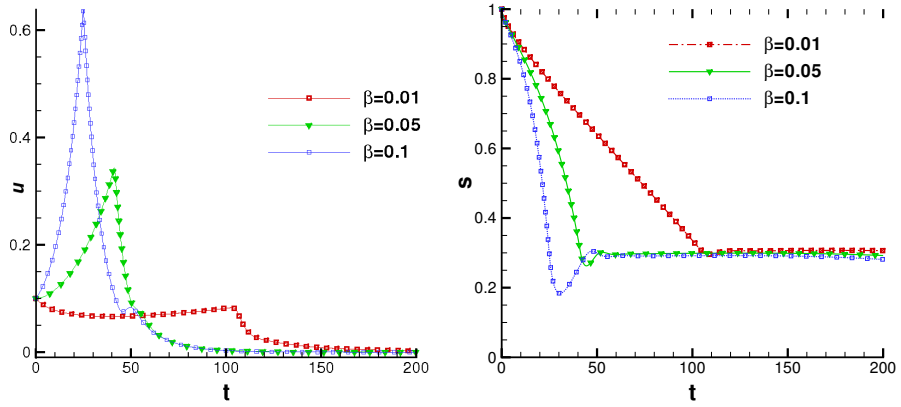


Figure 6.2: Effects of β on tumor cells density "u" (left) and oxygen "s" (right).

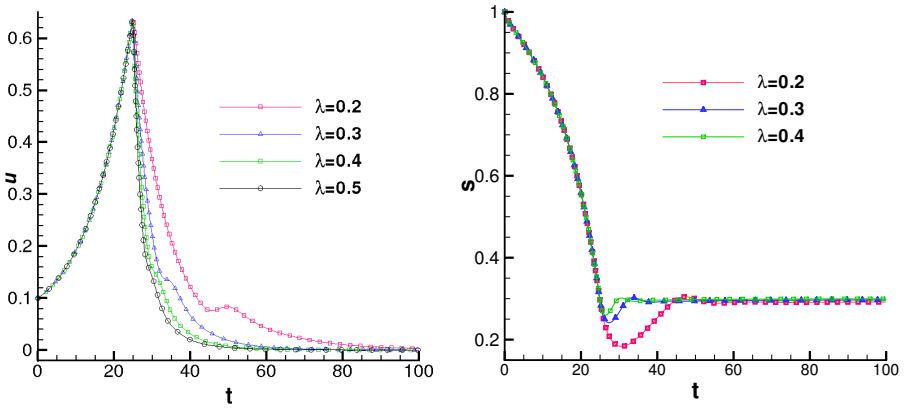


Figure 6.3: Effects of λ on tumor cells density "u" (left) and oxygen "s" (right).

(right) that μ effects directly on the oxygen concentration.

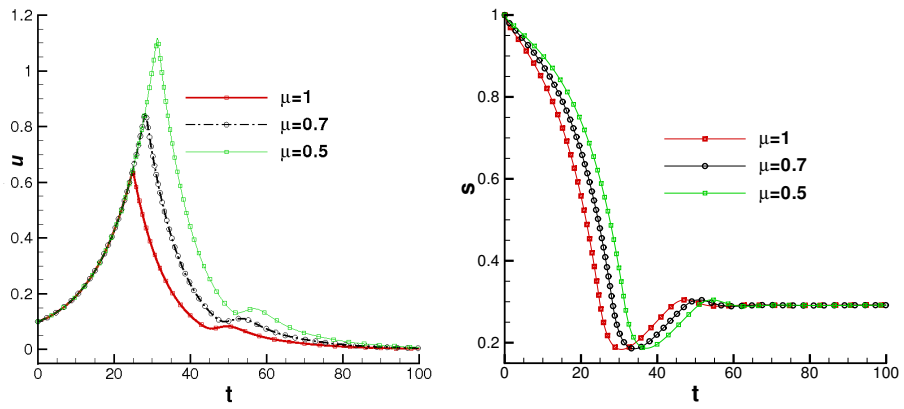


Figure 6.4: Effects of μ on tumor cells density "u" (left) and oxygen "s" (right).

Finally, we have calculated the hypoxia region by using different values of decaying factor of oxygen and studied its effect on tumor morphology and growth pattern at

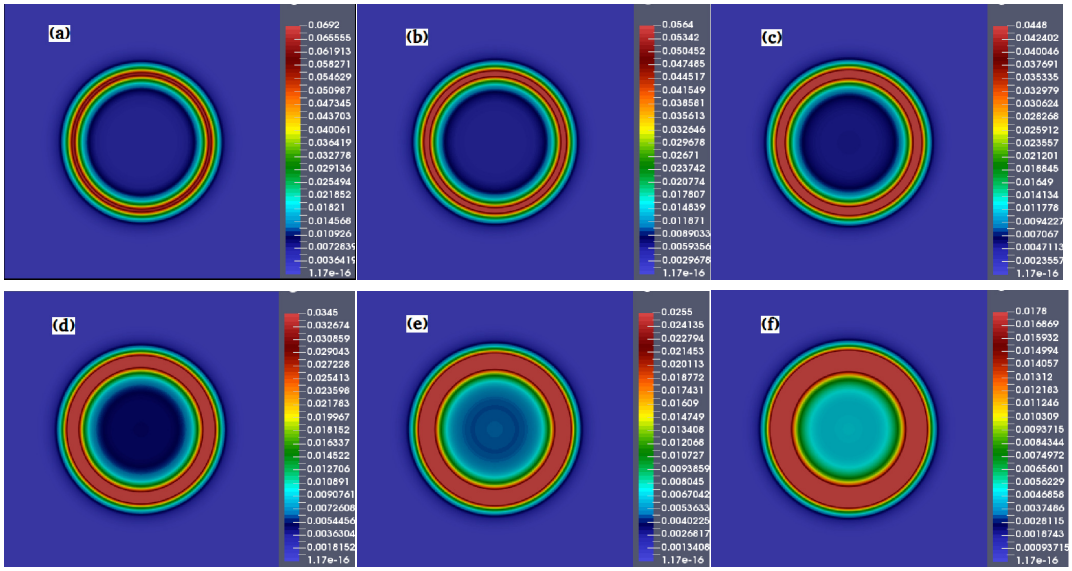


Figure 6.5: Tumor hypoxic and necrosis for; (a) $\rho = 0.000$, (b) $\rho = 0.002$, (c) $\rho = 0.004$, (d) $\rho = 0.006$, (e) $\rho = 0.008$ and (f) $\rho = 0.01$.

$t \approx 200$. These results demonstrates causes of tumor growth as cells propagation and cells dynamics. In figure 6.5(a-f), color scaling is different but the low density of live tumor cells's region is called hypoxia region and, central region and its surrounding called necrotic region. It also illustrate that how tumor cells adapt hypoxia by changing its nutrients (oxygen). From figures 6.5 $(a-f)$, it is shown that by increasing the decay factor of oxygen $\rho = 0 \rightarrow 0.01$, the maximum density of live tumor cells u decreases from $0.692 \rightarrow 0.0178$ and as a result, the tumor hypoxic region expands. Therefore, hypoxia region in the figure 6.5 (f) is larger as compared to figure 6.5 (b). Necrosis is a situation in which the tumor cells have lost oxygen. Figure 6.5 $(a-f)$ also demonstrates the tumor necrosis with different values of oxygen decaying parameter ρ. As we increase the value of $\rho = 0 \rightarrow 0.01$, the growth of live cells is less, so live cells movement is slow which results tumor necrosis region reduction.

6.6 Summary

In this work, we have proposed a new version of mathematical model for tumor growth by including the effects of natural decay of oxygen. The well-posedness of the governing partial differential equations is demonstrated. The resulting system of time dependent nonlinear partial differential equations are solved numerically by using finite element method. First, we have validated our computed results by assuming the decay

parameter $\rho = 0.0$ and good agreement is observed with reference solution. Next, we have studied the effects of oxygen decay parameter on density of tumor cells and have calculated the maximum age of growing tumor cells. It is observed that by increasing the value of decay parameter the density of live tumor cells is decreasing. From 2-D results it has been shown that the proposed mathematical model is able to predict the variations in hypoxic and necrotic cores. This work is a step forward for developing more realistic models for tumor growth and can be used to predict the response of tumor hypoxia in cancer treatment.

Chapter 7

Conclusions and Future Research Perspectives

This chapter summarizes the work done with some possible research directions in future. We have developed some numerical algorithms for solving nonlinear and stiff partial differential equations and proposed an efficient time discretization scheme along with conforming finite element space discretization. The algorithm is applied to numerically solve the phase-field models with focus on computational efficiency and robustness using DUNE-PDELab. We have also investigated parametric effects of the physical phenomena used in this work based on phase-field models.

In chapter 1, we have provided an overview of phase-field models, their formulations and some of their applications. Additionally, challenges and difficulties in numerical simulation of phase-field model are provided. The detailed numerical discretization both in time and space are given in chapter 3 and have briefly described the implementation of these schemes. In chapter 3, we numerically solved the 2nd Allen-Cahn equation and have provided error estimates of the method. Solutions to some benchmark problems are computed and comparison of the method is provided. Chapter 4 provides an extension of the work done in chapter 3 by numerically solving the 4th Cahn-Hilliard equation having variable mobility and simulated the spinodal decomposition phenomena. In chapter 5, we have solved the coupled system of two partial differential equations known as the dendritic crystal growth model. The effect of parameters evolving in mathematical model have been studied. In chapter 6, we focused on mathematical formulation and simulation of tumor growth model. We have reviewed some oxygen dependent tumor growth and observed that solid tumor includes the area where oxygen concentration is very low called hypoxia, often in the surrounding areas of necrosis. Hypoxic cells in these areas are resistant to chemotherapy and radiation effect.

In future, we intend to extend this work using adaptive time-space algorithm with the help of moving grids strategy to simulate some more realistic 3-D free boundary problems like wound healing and flood flows etc. We are already working on developing of an algorithm for coupled incompressible Navier-Stokes/Cahn-Hilliard equations to study multi-phase flow problems.

Chapter 8

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